WEBNOMAD STUDIO · WEBNOMAD.ORG

Retirement Income Planner

User Guide

Version 7

The first retirement planner you can chat with — that teaches you as you plan, verifies its own AI suggestions, and stays accurate for years.

Published: June 2026 · Supersedes the v6 User Guide
WebNomad Studio · webnomad.org · dev@webnomad.org

How to use this guide

This guide walks you through every feature of the Retirement Income Planner, version 7 — from opening the file for the first time to chatting with the new **Plan with AI** co-pilot and running AI audits of your help content. It assumes no prior knowledge of retirement-planning jargon: terms like MAGI, RMD, ACA cliff and IRMAA are explained the first time they appear.

You can read the guide front-to-back to learn the planner systematically, or jump to a chapter from the Table of Contents.

Two example users

Most chapters use one of two example users so screenshots tell a coherent story:

- **Linda** — a US single filer retiring at 62 with $400,000 in her 401(k), $50,000 in cash and $80,000 in equity. She plans to claim Social Security at 67. Most US-tax, ACA and IRMAA chapters follow Linda's plan.
- **David & Jo** — a UK retired couple living in the Philippines on UK State Pension plus a small US Social Security entitlement. They appear in the multi-currency, foreign-resident and treaty chapters.

Conventions

- **Bold** marks button labels and tab names.
- *Italics* mark a phrase you'll see displayed in the planner.
- Numbered procedures (1, 2, 3...) are step-by-step actions — follow them in order.

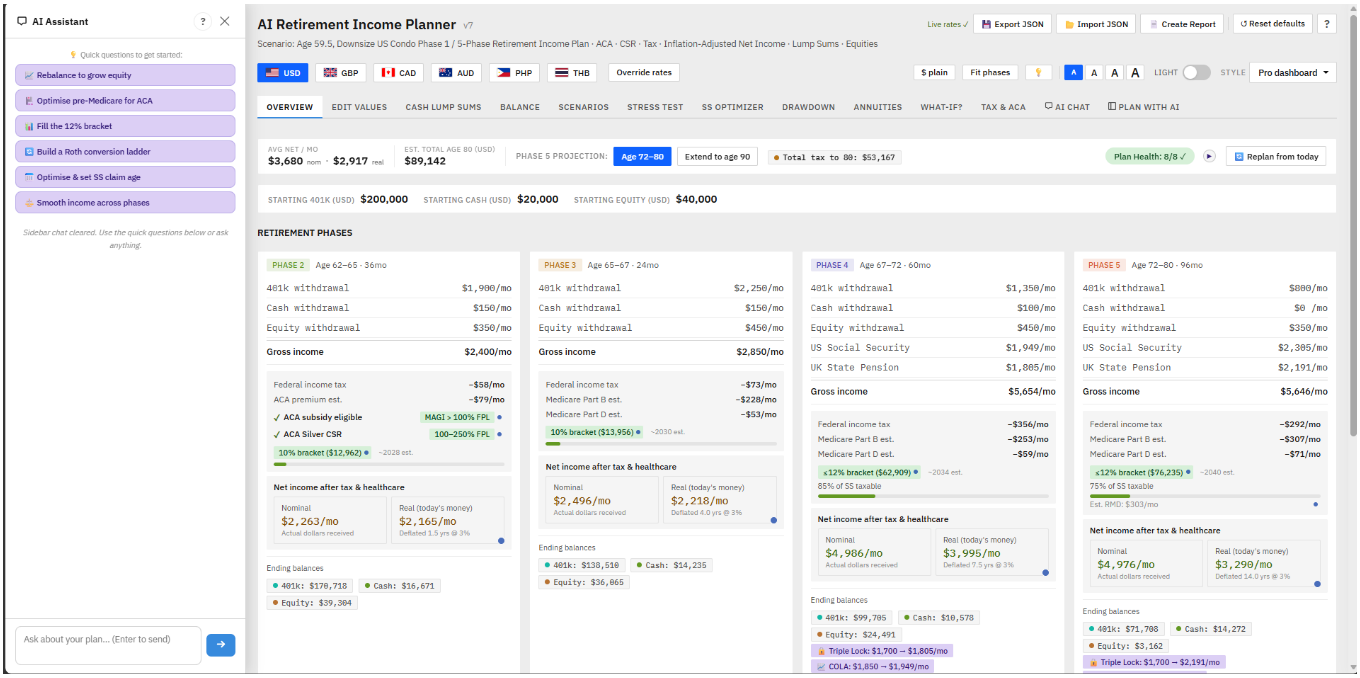
Disclaimer

This guide and the Retirement Income Planner are educational tools for personal planning. They do not constitute financial, tax, legal or investment advice. All projections are estimates based on the values you enter — not guarantees. Tax law, ACA rules, Medicare premiums, Social Security structures and overseas tax parameters change frequently. The planner has tools to help you keep its figures and rules current (see Chapter 37), but you are responsible for verifying your plan reflects current law and your personal circumstances. Always consult a qualified financial advisor, tax professional or retirement planner before making any financial decisions.

What's new in version 7

Version 7 builds on v6's AI advisor, education system and longevity tools. If you are upgrading, here is everything that changed — every existing plan loads unchanged.

Area	What's new in v7	Chapter
Plan with AI co-pilot	A dockable AI sidebar you can open from any tab. It updates your plan in plain English, **tests its own suggestions** in the simulator before you apply them, auto-refines proposals that would run you out of money, and gives you one-click **Undo**.	20
Part-time income	Now available in **every** phase (was phases 1–2 only).	7
State income tax	Type-ahead state picker plus an **AI auto-fill** that looks up your state's flat rate and exemptions.	7, 10, 21
Enriched PDF report	Adds an asset-allocation breakdown, a plan-resilience page (Monte Carlo + the 12-scenario stress grid) and portfolio charts. Blank-page printing bug fixed.	23, 35
AI Chat	A **copy** icon on every AI response for pasting into Word or email.	19
Tabs & layout	Tabs reordered into a left-to-right workflow; a polished sticky header (lift shadow, frosted bar, sliding tab marker); grouped Overview toolbars.	33
Themes	Two new visual styles — **Pro dashboard** and **High contrast** (accessibility) — for five styles in total.	33
Education	New ⓘ popovers (getting-started, a distinct 22%-bracket explainer, a dedicated SS-claiming-age explainer, cash/bond volatility); survivor-scenario setup guidance.	25, 28
Feature Tour	Expanded from 7 to **9 stops** — now includes Plan with AI and Create Report.	31
Historical backtest NEW	Replays your plan against the actual sequence of US market returns for every overlapping retirement window since 1928 — a sequence-of-returns confidence check alongside Monte Carlo. On the Stress test tab.	14
Cash lump sums — money out	Lump sums now work both ways: one-off **expenses** (roof, car, medical) as well as windfalls, with a per-row Money in / Money out selector.	8
Confidence summary NEW	Plan Health now opens with a plain-English summary uniting all three confidence lenses (checks, Monte Carlo, backtest) and a spending-headroom line.	11
Maximum sustainable spending NEW	A transparent "how much can I safely spend?" finder on the What-if? tab, with solvent / Monte Carlo / historical gates.	25
Education	New **retirement spending smile** concept (go-go / slow-go / no-go) with a Concepts-to-learn prompt and glossary entry.	30, 31



Version 7: the Plan with AI co-pilot docks on the left while your dashboard stays fully visible on the right.

Table of contents

Part 1 — Getting started

1. What this planner does (and what it doesn't)
2. Opening the file for the first time
3. The Concepts Primer: 5 ideas worth knowing
4. The Setup Wizard: your first plan in 3 minutes
5. Already retired? The Replan-from-today path

Part 2 — Building your plan

6. Reading a phase card
7. The Edit tab: where the numbers live
8. Cash lump sums (money in and money out)
9. Currencies and residency modes
10. Tax parameters: what each one means

Part 3 — Understanding your results

11. Plan Health: the 8 checks and the score
12. Acknowledging checks that don't apply
13. The five drawdown strategies
14. Stress Test (and the historical backtest)
15. Monte Carlo simulation
16. Scenarios comparison
17. Balance chart

Part 4 — AI features

18. Adding your Anthropic API key
19. AI Chat: six starter prompts and how to read proposals
20. Plan with AI — the co-pilot sidebar NEW
21. Fetch current rates (and state-tax lookup) 22. Help content audit and one-click corrections
23. The PDF report (now with allocation, resilience & charts)

Part 5 — Specialized tools

24. Annuity Calculator
25. What-if Explorer (and the survivor scenario)
26. Social Security Optimizer
27. Roth Conversion Optimizer integration

Part 6 — Learn as you plan

28. ⓘ icons: click any field you don't understand
29. Clickable phase-card badges
30. Concepts-to-Learn panel
31. Glossary and the 9-step Feature Tour
32. Tax & ACA notes tab

Part 7 — Settings and maintenance

33. Text size, dark mode, themes & the polished header
34. Saving, exporting, importing
35. PDF export
36. Keeping your planner current
37. Troubleshooting and FAQ

Appendices

A. Keyboard shortcuts   B. JSON export field reference   C. Where to learn more

Part 1 — Getting started

The planner is a single HTML file that runs entirely in your browser. There is no installation, no account, no subscription, and your data never leaves your computer. This part walks you through your first 15 minutes with the planner.

Chapter 1 — What this planner does (and what it doesn't)

What it does

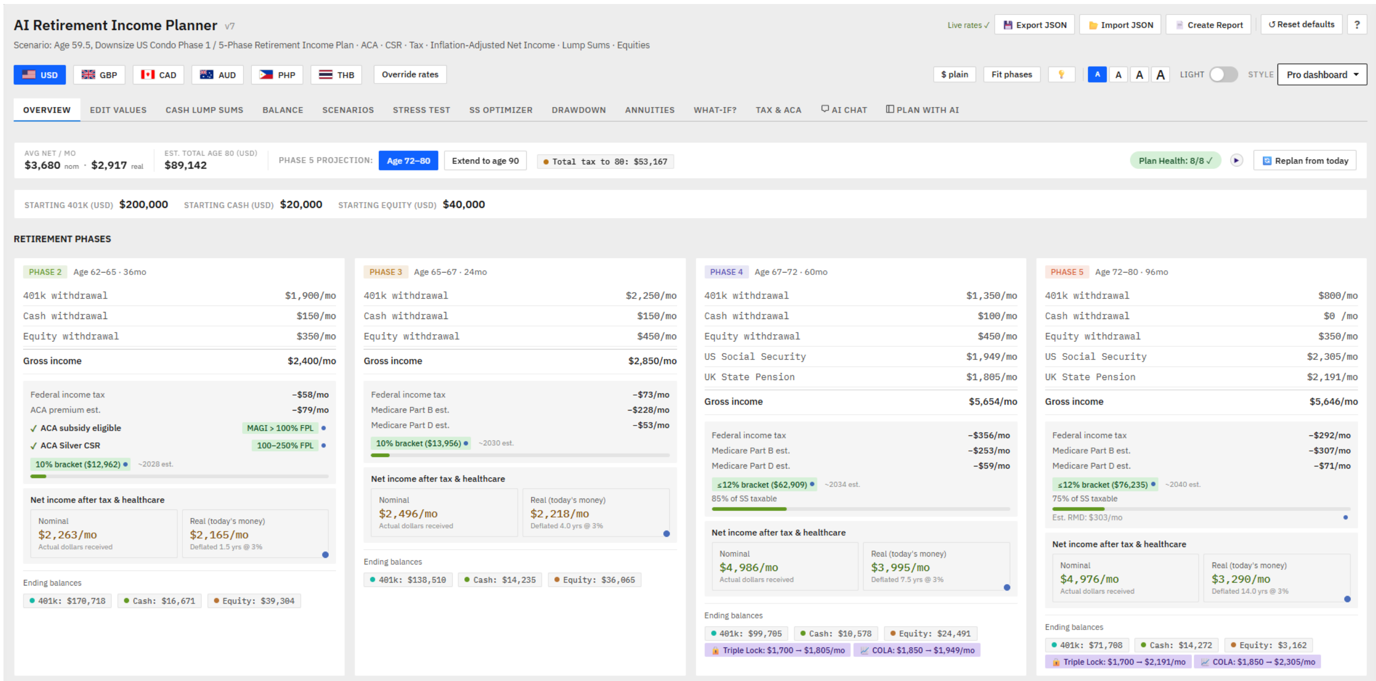
The Retirement Income Planner models your retirement, month by month, across up to five distinct phases. You tell it what you have today (401k, cash, equity, Roth, pensions), what you'll draw in each phase, and what tax rules apply. It returns:

- Your net monthly income in every phase, in nominal and real (today's-money) terms.
- Your account balances at the start and end of each phase.
- Your effective tax rate, marginal bracket, ACA subsidy status, IRMAA risk and Required Minimum Distribution exposure.
- A Plan Health score against eight common retirement risks.
- Five drawdown-strategy comparisons plus a Monte Carlo simulation.
- A built-in AI advisor that knows your plan — available both as a full-screen **AI Chat** tab and, new in v7, as a dockable **Plan with AI** co-pilot sidebar that updates your plan in plain English and verifies its own suggestions before you apply them.
- A clean, multi-page **PDF report** — now including an asset-allocation breakdown, a plan-resilience page and portfolio charts.

What it doesn't do

- It is **not** financial advice. The math is accurate; what you do with it is your decision.
- It does not connect to your bank, brokerage or HMRC/IRS. You enter the numbers, and they stay on your computer.
- It does not predict markets. Returns and inflation are assumptions you set.
- It does not model every product. Variable annuities, complex options strategies and bespoke tax structures are out of scope.

> **Why a single HTML file?**
> Privacy: your plan never leaves your machine. Longevity: an HTML file works in any browser, for years, without a hosted service. Portability: copy it to a USB stick, email it to your spouse, or share it with your advisor — it's just one file.



The Overview dashboard: your whole retirement, phase by phase, at a glance.

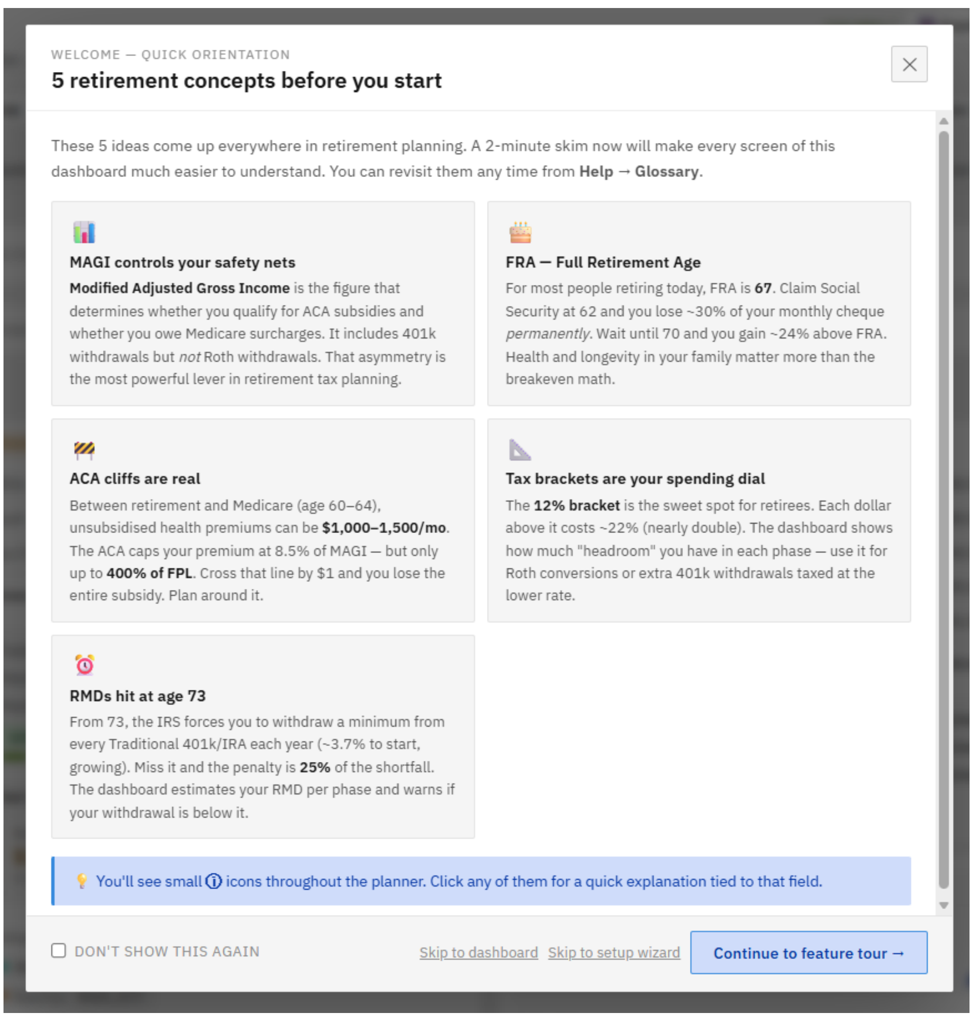
Chapter 2 — Opening the file for the first time

You'll have received the planner as a download after purchase — a single .html file like Retirement-Planner-v7-v272.html. The number at the end is the build; it changes with each release, so yours may differ.

To open the planner:

1. Save the `.html` file to a folder on your computer (your Documents folder is fine).
2. Double-click the file. It opens in your default browser (Chrome, Edge, Firefox or Safari all work). 3. On first open, the **Concepts Primer** and **Setup Wizard** appear automatically (next two chapters).

> **Tip — make a desktop shortcut**
> Right-click the file and choose *Send to → Desktop (create shortcut)* on Windows, or drag it to your Dock on macOS. Your plan auto-saves to the browser's local storage on every change — re-open in the same browser and your plan is still there.



On first open, a five-card primer introduces the ideas that drive most retirement decisions.

Chapter 3 — The Concepts Primer: 5 ideas worth knowing

Before you build your first plan, the planner shows a short primer covering five ideas that drive most decisions. Skip it on your first visit if you like — you can relaunch it any time from the Help modal.

The five concepts

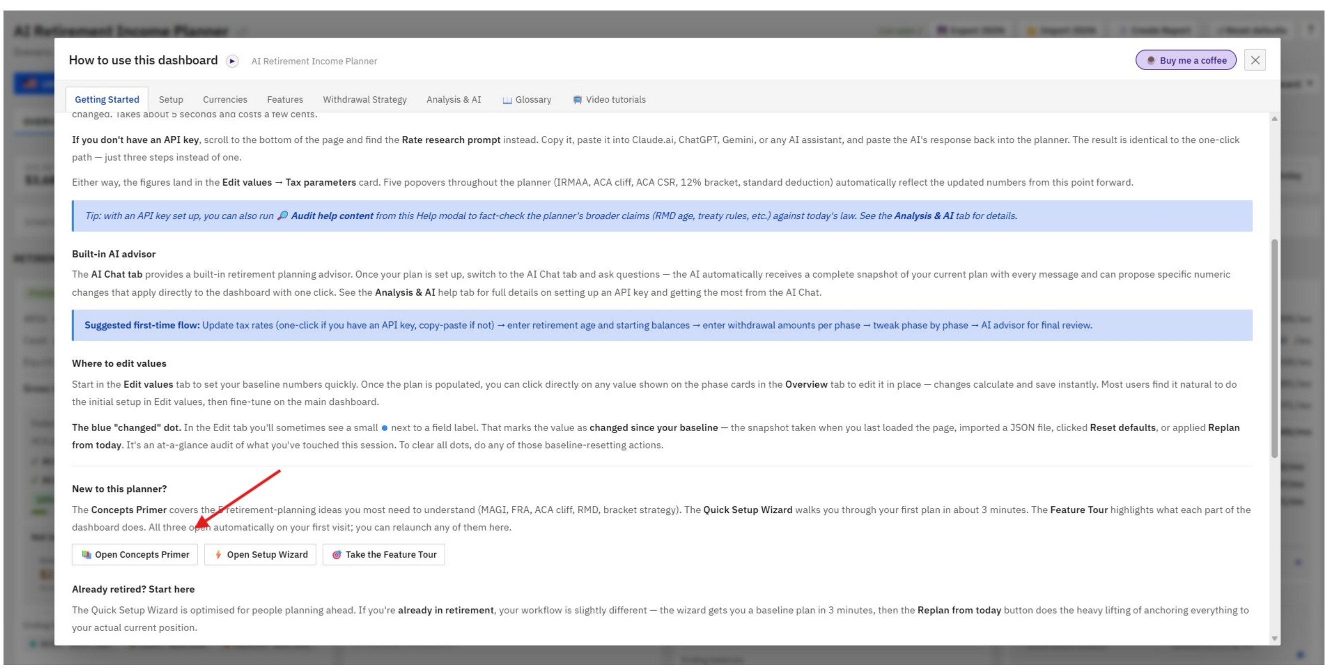
1. MAGI controls your safety nets. Modified Adjusted Gross Income determines whether you qualify for ACA subsidies and whether you owe Medicare surcharges (IRMAA). It includes 401k withdrawals but *not* Roth withdrawals — that asymmetry is the most powerful lever in retirement tax planning.

2. FRA — Full Retirement Age. For most people retiring today, FRA is 67. Claim at 62 and you lose about 30% permanently; wait to 70 and you gain about 24% above FRA.

3. ACA cliffs are real. Between retirement and Medicare, unsubsidised premiums can be $1,000–1,500/month. The ACA caps your premium at 8.5% of MAGI — but only up to 400% of the Federal Poverty Level. Cross by $1 and you lose the entire subsidy.

4. Tax brackets are your spending dial. The 12% bracket is the sweet spot; each dollar above it costs about 22%. The planner shows your *headroom* in each phase.

5. RMDs hit at age 73. From 73 the IRS forces a minimum withdrawal from Traditional 401k/IRA each year. Miss it and the penalty is 25% of the shortfall.



Reopen the primer any time from Help → Getting Started.

Chapter 4 — The Setup Wizard: your first plan in 3 minutes

After the Primer (or any time from the Help modal), the Setup Wizard guides you through the minimum inputs needed for a working plan.

The wizard's five steps

1. Currency & residency. Choose your home currency (USD, GBP, CAD, AUD, PHP or THB) and tax residency. This determines which tax regime applies and which Plan Health checks are relevant.

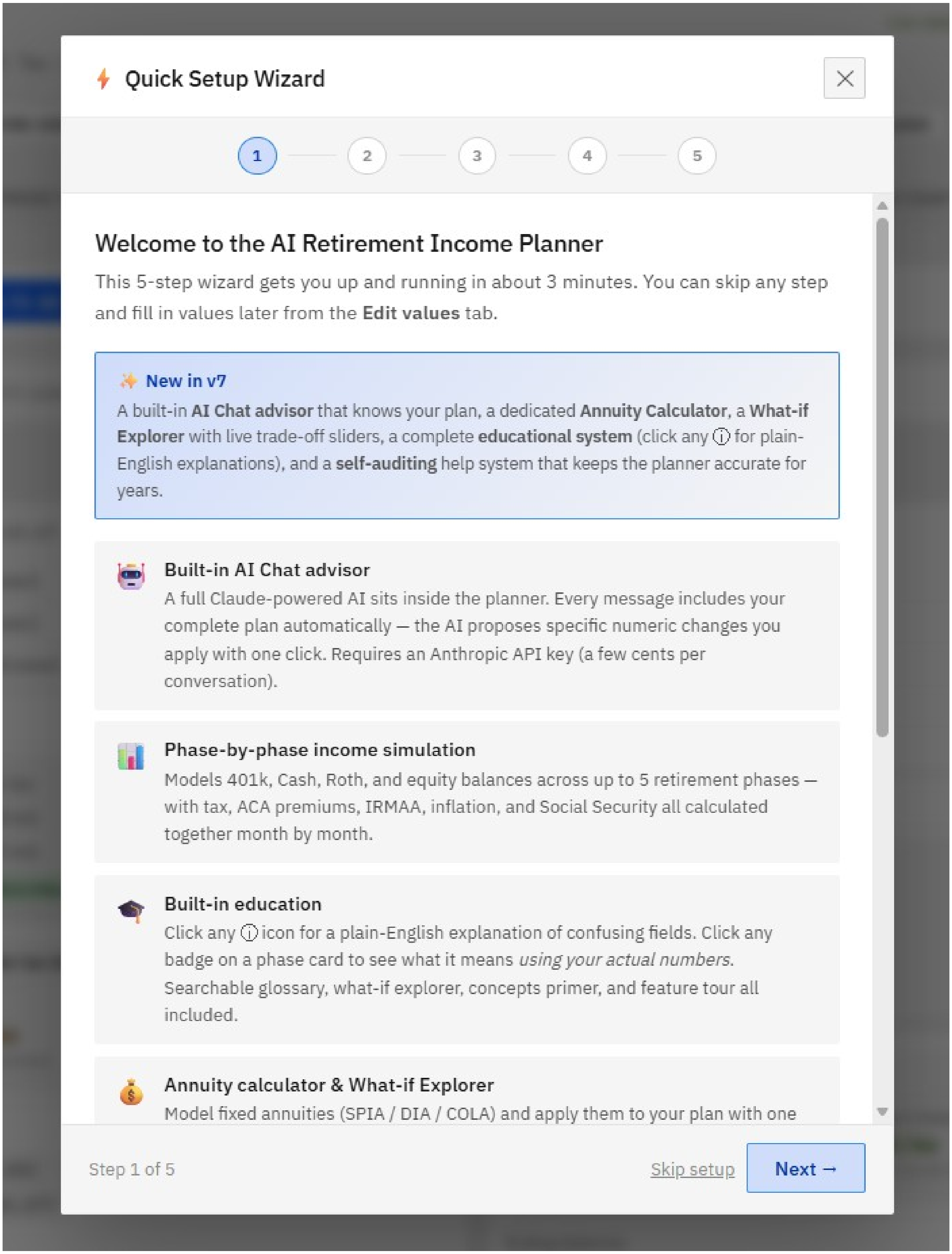
2. Starting balances. Enter 401k, Cash, Equity and Roth balances today. You can refine later.

3. Retirement start age & plan horizon. Choose the age you stop working and the age your plan runs to. The default is 80; most users extend to 90 for longevity protection using the **Extend to age 90** preset on the command strip.

4. Social Security / State Pension. Enter your expected monthly benefit at your chosen claiming age. UK users enter their State Pension forecast.

5. Tax parameters. The wizard ships with current-year defaults. With an Anthropic API key, one click updates these via **Fetch current rates** (Chapter 21).

> **After the wizard**
>
> You'll land on the Overview dashboard. Fine-tune any input on the **Edit values** tab or by clicking values inline on the phase cards. The Plan Health badge immediately shows whether your plan passes the eight standard checks — don't worry if it's not 8/8 (see Chapters 11–12).



The five-step wizard builds a working plan in a few minutes.

Chapter 5 — Already retired? The Replan-from-today path

Most planners assume you're modelling forward from accumulation. If you're already retired, that's the wrong starting point. **Replan from today** is designed for you.

How it differs from the normal wizard

The wizard asks "*at what age will you retire?*" Replan-from-today asks "*how old are you now, and what are your actual current balances?*" It then:

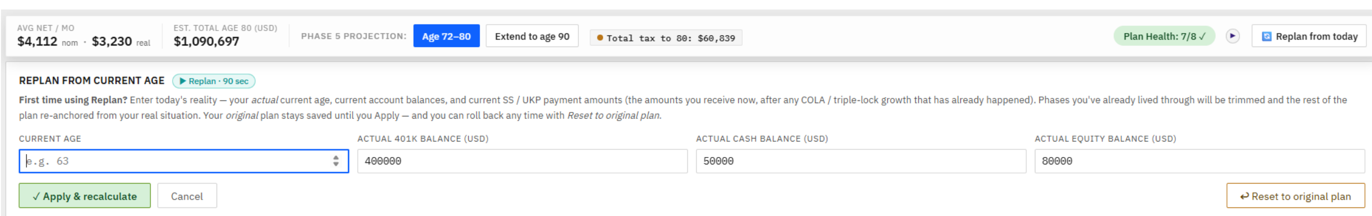
- Trims past phases out of the model so they don't double-count.
- Pre-fills your current Social Security and UK State Pension amounts.
- Preserves your spouse's age gap if you file MFJ.
- Restarts the projection from your actual current balances.

Recommended flow

1. Step through the Setup Wizard with your real retirement start age.
2. On the **Edit values** tab, click **Fetch current rates** (or use the Rate Research Prompt). 3.

Click **Replan from today** and enter your current age and actual balances.

4. Open the **Plan with AI** sidebar (Chapter 20) and ask for optimisations from your anchored position.



Replan from today anchors an already-retired plan to your real current age and balances.

Part 2 — Building your plan

With your first plan loaded, this part walks through the day-to-day mechanics of inspecting and editing your numbers.

Chapter 6 — Reading a phase card

The **Overview** tab shows your retirement as a strip of phase cards (typically five). For Linda these are roughly ages 62–65, 65–67, 67–72, 72–80, and 80+.

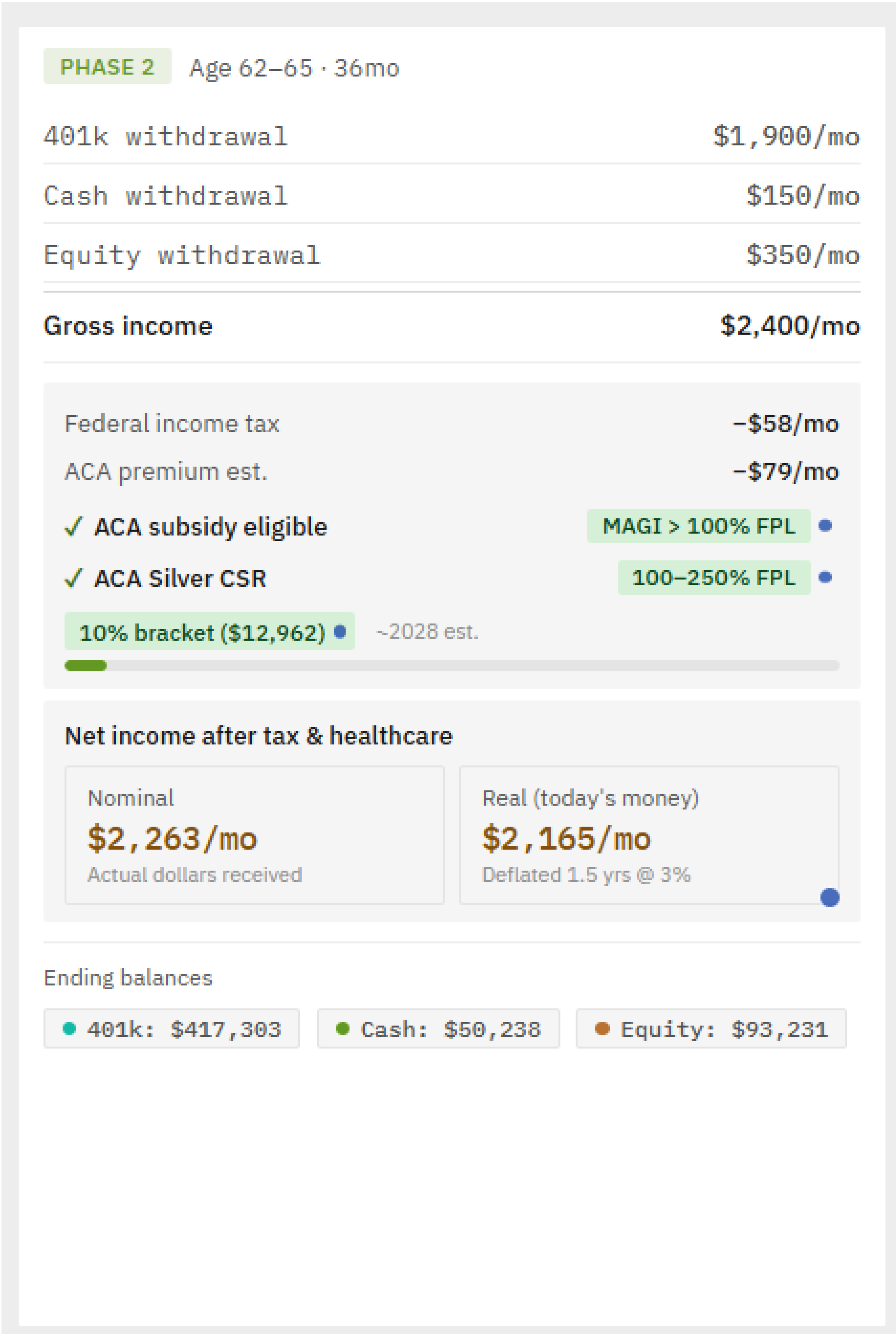
Anatomy of a phase card

- **Age range and label** — e.g. *Phase 2 (ages 65–67)*.
- **Net monthly income** — what you'll receive after tax, in nominal and real terms.
- **Income source breakdown** — colour-coded segments for 401k, cash, equity, Roth, Social Security, pensions, part-time work and rental income.
- **Tax & healthcare bar** — federal tax, ACA premium or Medicare cost, and IRMAA surcharge.
- **Account balances** — start and end balance per account with direction arrows.
- **Health badges** — clickable badges flagging conditions like *Near 12% bracket ceiling* or *Below RMD estimate*.

> **New in v7 — clearer phase cards**
> The *"% of SS taxable"* note now always sits on its own line beneath the bracket badge, so it never crowds the bracket pill at narrow widths. The Roth withdrawal and conversion sections now show each phase's duration in months (e.g. *Phase 5 · 96mo*) so you can see the period a figure covers at a glance.

Clickable badges

Every badge is clickable, and the explanation uses your actual figures — for instance: *"You're $4,200 below the 12% ceiling — you could withdraw an extra $4,200 from your 401k or convert that much to Roth without bumping into 22%."*



Every element of a phase card — and every badge is clickable for an explanation in your own numbers.

Chapter 7 — The Edit values tab: where the numbers live

Click the **Edit values** tab to see every input that drives your plan, organized into cards by topic.

The cards

- **Starting balances** — 401k, Cash, Equity, Roth today, plus each one's annualized growth rate.
- **Per-phase withdrawals** — for each phase, what you withdraw from each account, plus part-time income, Roth conversions, rental/passive income and Social Security.
- **Social Security & pensions** — base amount, COLA rate, claiming age, spouse details (if MFJ).
- **Tax parameters** — federal brackets, FPL thresholds, IRMAA cliff, standard deduction, and **state income tax** (see below). See Chapter 10.
- **Country tax regimes** — UK personal allowance and bands, Canadian brackets, Australian levy and brackets.
- **RMD settings** — start age (default 73) and whether RMD warnings are active.
- **Monte Carlo settings** — number of simulations and volatility (sigma) per account type. It also holds **Backtest growth-bucket stock %** — the stock/bond split (default 60/40) used by the Historical backtest on the Stress test tab (Chapter 14).

> **New in v7 — part-time income in every phase**
> Earlier versions offered a part-time income field only in the first couple of phases. In v7 **every phase** has its own part-time income field, so you can model a phased "glide-path" retirement (e.g. consulting two days a week until 70) accurately. The field flows through every calculation, the Plan Health checks, the AI context and the PDF report.

> **New in v7 — state income tax with autocomplete + AI lookup**
> The **State** field offers a type-ahead list of US states as you type. Once a state is entered, an **Auto-fill rate** button becomes active: click it and the AI looks up that state's flat-rate approximation and any retirement-income exemptions and fills them in for you. Clear the state and the button disables again. (Requires an API key — see Chapter 18.)

Inline editing & blue-dot indicators

You can change any number anywhere. Edit-tab inputs update the dashboard live, and you can click directly on dashboard phase-card numbers to edit in place (USD display mode only). A small blue dot marks any field changed since the session started; save or import a plan to reset the baseline.

PHASE 3 Age 65–67 · 24mo

401k withdrawal	$2,250/mo
Cash withdrawal	$150/mo
Equity withdrawal	$450/mo
Part-time work	$1,200/yr
Gross income	**$2,950/mo**

Federal income tax	−$83/mo
Medicare Part B est.	−$228/mo
Medicare Part D est.	−$53/mo

10% bracket ($13,956) ~2030 est.

Net income after tax & healthcare

Nominal	Real (today's money)
$2,586/mo	$2,298/mo
Actual dollars received	Deflated 4.0 yrs @ 3%

Ending balances

401k: $422,035 · Cash: $50,412 · Equity: $101,877

v7 adds part-time income to every phase and an AI-powered state-tax lookup.

Chapter 8 — Cash lump sums (money in and money out)

Real retirements include one-time cash events — selling a house, an inheritance, a deferred bonus, but also one-off *expenses* like a new roof, a car, a wedding, or a big medical year. The **Cash lump sums** tab lets you add both kinds without distorting your phase withdrawal rates.

Adding a cash event

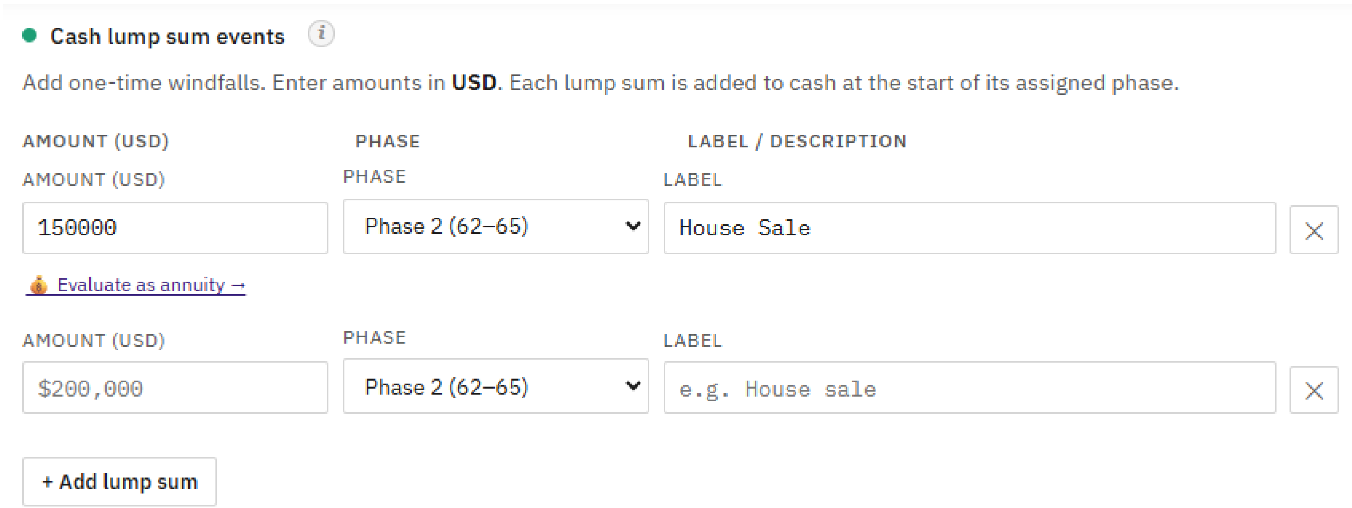
1. Click the **Cash lump sums** tab.
2. Click **+ Add cash event**.
3. Set its **Type** — *Money in* (a windfall) or *Money out* (a one-off expense).
4. Enter the amount, the phase it lands in, and an optional label (e.g. *House sale* or *New roof*).

Money in is added to your cash balance at the start of that phase and compounds at your cash growth rate. **Money out** is taken from cash first, then cascades to your other accounts (equity → 401k → Roth → Super) if cash runs short.

> **Money in adds funds — it doesn't spend them**
> A *money-in* lump increases your cash balance but doesn't automatically increase what you spend. To use it for living expenses, raise your cash withdrawal in the same or a later phase on the Edit values tab.

> **Large 401k-funded expenses**
> A *money-out* event is treated as a straight cash event with no extra income tax applied. If a big expense would really be funded from your 401k, model it instead by raising that phase's 401k withdrawal, so the income tax is captured accurately.

On the phase cards, a *money-in* event shows as a green "*+ $X lump sum added*" pill and a *money-out* event as a neutral "*− $X one-time expense*" pill. For *money-in* rows with an amount, an **Evaluate as annuity** link appears, switching to the Annuity Calculator with the lump's value pre-filled as the premium (see Chapter 24).



Model money in and money out — one-off windfalls and one-off expenses, each in the right phase.

Chapter 9 — Currencies and residency modes

The planner supports six currencies and four residency modes. The combination determines which tax regimes apply and which health-care assumptions are used.

Switching currency & the two currency flags

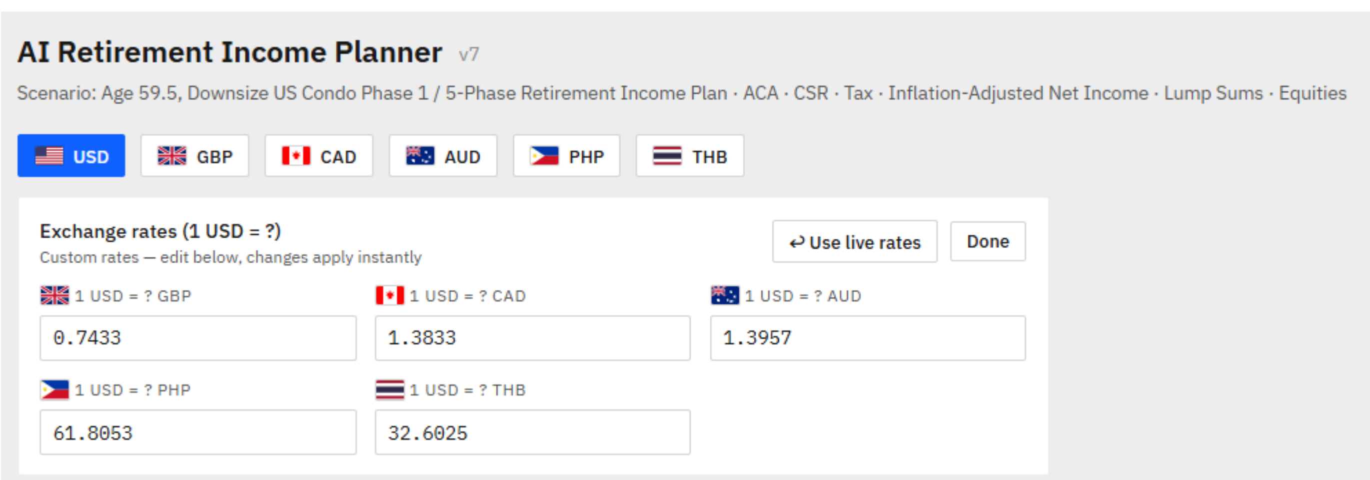
Use the currency selector in the header (USD / GBP / CAD / AUD / PHP / THB). All values are stored internally in USD; the display converts using live exchange rates. Two flags control behaviour: *active currency* (your primary residency currency) and *display currency* (what's shown). You can browse a plan in one currency while modelling it in another.

Residency modes

- **US resident (USD)** — full US federal tax, IRMAA, ACA subsidy and Medicare modelling.
- **UK resident (GBP)** — UK income tax on 401k withdrawals, State Pension, equity gains and part-time income; US Social Security is taxed in the UK, not the US, under Article 17(3) of the US-UK treaty, which makes social security taxable only in the country you live in. US tax is reduced by a Foreign Tax Credit.
- **Foreign resident (PHP / THB / other)** — disables US healthcare costs (ACA, Medicare, IRMAA). US federal tax still applies if you remain a US person.
- **Canada / Australia** — full Canadian or Australian tax regime including CPP/OAS or Super/Age Pension.

New in v7 — close the custom-rates panel without losing your rates

If you enter your own exchange rates, you can now click **Done** to close the custom-rates panel while *keeping* those rates active — the reopen control is relabelled *Edit custom rates*. A separate *Use live rates* button discards your custom values and refetches live rates.



Set your own exchange rates, then click Done to keep them active while you keep planning.

Chapter 10 — Tax parameters: what each one means

On the Edit values tab, the **Tax parameters** card holds the values that drive every tax calculation. Every value is inflation-adjusted forward to each phase's mid-point automatically.

The federal parameters (US)

- **Standard deduction** — single (and MFJ if filing jointly), plus an extra senior deduction for filers 65+.
- **Tax bracket ceilings** — taxable-income thresholds for the 10%, 12% and 22% brackets; above 22% the model applies 24%.
- **IRMAA threshold** — the single-filer MAGI level above which IRMAA Tier 1 surcharges apply (doubled for MFJ).
- **FPL 100% / 250% / 400%** — Federal Poverty Level values; the ACA marketplace uses the prior year's FPL.
- **State income tax** v7 — a flat-rate approximation plus exemptions for your state, with type-ahead entry and optional AI auto-fill (Chapter 7).

UK / Canadian / Australian parameters

UK: personal allowance, 20% and 40% ceilings, 45% additional-rate threshold (USD-equivalent, auto-converted). Canadian: basic personal amount, four bracket ceilings plus average provincial rate. Australian: tax-free threshold, three bracket ceilings, the four marginal rates themselves, and the Medicare levy with its income threshold.

> **Keeping these current**
> Use **Fetch current rates** (Chapter 21) for one-click updates with an API key, or the **Rate Research Prompt** for any external AI. See Chapter 36 for the long-term workflow.

TAX PARAMETERS (2026 USD)

Current-year US federal thresholds. The dashboard automatically inflates these for each future phase using your inflation rate. Click any ⓘ for a plain-English explanation of why each field matters. To refresh these values for the current tax year, use the **Rate research prompt** at the bottom of this page.

STANDARD DEDUCTION — SINGLE ($) ⓘ

16100

AGE 65+ ADDITIONAL DEDUCTION ($) ⓘ

2050

10% BRACKET CEILING (TAXABLE INCOME $) ⓘ

12400

12% BRACKET CEILING (TAXABLE INCOME $) ⓘ

50400

22% BRACKET CEILING (TAXABLE INCOME $) ⓘ

105700

IRMAA THRESHOLD — SINGLE ($) ⓘ

109000

NIIT MAGI THRESHOLD — SINGLE ($) ⓘ

200000

3.8% surtax on investment income above this MAGI (MFJ uses $250k). Not inflation-indexed in law, but the model inflates it per phase like other thresholds.

ACA 400% FPL CLIFF — SINGLE ($) ⓘ

62600

ACA 250% FPL — SILVER CSR CEILING ($) ⓘ

39125

FEDERAL POVERTY LEVEL 100% — SINGLE ($) ⓘ

15060

Every tax parameter in one place — inflation-adjusted to each phase automatically.

Part 3 — Understanding your results

With your plan built, this part covers how to interpret the planner's outputs — from the Plan Health score to the drawdown strategies and the Monte Carlo fan chart.

Chapter 11 — Plan Health: the 8 checks and the score

Plan Health is a quick-glance score, shown as a badge on the dashboard (e.g. *6/8*). Click it to open the full panel.

The 8 checks

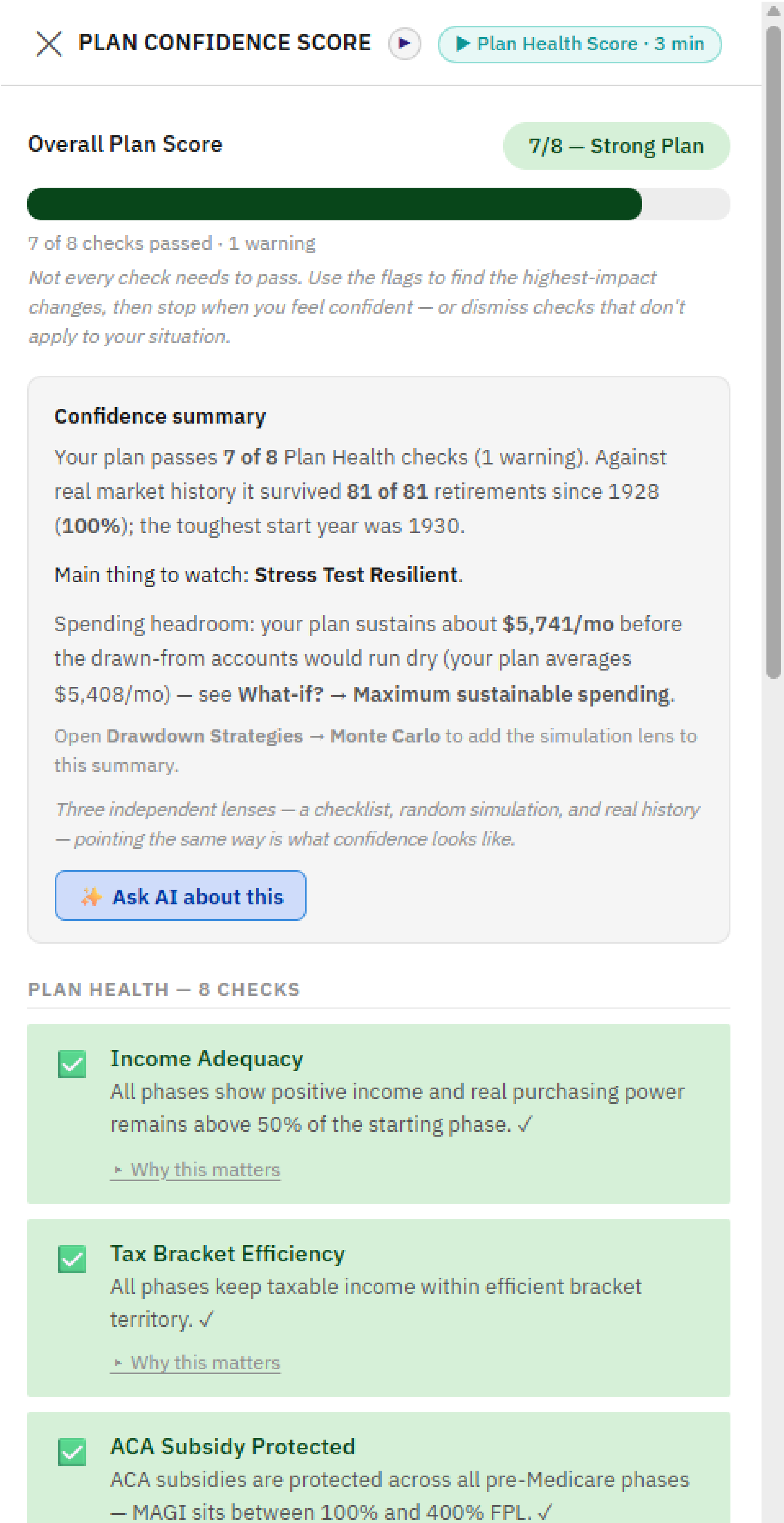
1. **Portfolio survives to plan end** — combined balances stay above zero through your final phase.
2. **Income Floor Foundation** — guaranteed income (SS, pensions) covers a meaningful share of your target.
3. **Tax bracket headroom** — you're not needlessly pushing into 22% with 12% headroom left.
4. **ACA Subsidy Protected** — pre-Medicare phases stay below the 400% FPL cliff.
5. **IRMAA cliff distance** — post-Medicare phases stay below the IRMAA Tier 1 threshold.
6. **RMD readiness** — post-73 phases withdraw at least the estimated RMD.
7. **Real income preserved** — inflation-adjusted income doesn't decline materially across phases.
8. **Cash buffer adequate** — you hold enough cash to weather a 2–3 year downturn.

The badge colour reflects the percentage passed: green ≥75%, amber ≥50%, red below. Each check has an expandable **Why this matters** explanation. Acknowledged checks add *(N acknowledged)* to the badge.

The Confidence summary NEW

At the top of the Plan Health panel, a short **Confidence summary** draws your three independent confidence lenses into one or two plain-English sentences — for example: "*Your plan passes 5 of 8 Plan Health checks (1 warning). In Monte Carlo it stays solvent in 92% of randomized runs to age 90. Against real market history it survived 78 of 78 retirements since 1928 (100%); the toughest start year was 1929.*" A closing line names the **main thing to watch** — your highest-impact failing check if you have one, otherwise the residual risk (typically a poor first decade of returns). It also shows a **spending-headroom** line — roughly how much your plan sustains before the drawn-from accounts would run dry — linking through to the What-if Maximum-sustainable-spending tool (Chapter 25).

> **Why three lenses matter**
> The summary deliberately combines a *checklist* (Plan Health), a *random simulation* (Monte Carlo) and a *replay of real history* (the backtest, Chapter 14). When all three point the same way, that agreement is a stronger signal of a robust plan than any single number. The Monte Carlo line appears only once you've run that simulation (Drawdown Strategies → Monte Carlo); until then the summary invites you to add it. An ✨ **Ask AI about this** button hands the whole summary to the AI and asks for the single most impactful change.



Eight checks plus a plain-English Confidence summary at the top — each check with a personalised status and a Why-this-matters explanation.

Chapter 12 — Acknowledging checks that don't apply

Some checks won't be relevant — an ACA cliff check is irrelevant for a foreign-resident plan; an RMD check is irrelevant if your plan ends before 73. Rather than living with a permanent red flag, acknowledge the check.

1. Open the Plan Health panel.
2. Find the check you want to acknowledge.
3. Click **I'm OK with this** beneath its description.
4. Optionally enter a short reason (for your own records).
5. Click **Acknowledge**.

The check stays in the panel but faded, with a **Restore** link. Acknowledgments persist across reloads and travel with your plan when you export JSON.

PLAN CONFIDENCE SCORE ▶ Plan Health Score · 3 min

ACA Subsidy Protected

ACA subsidies are protected across all pre-Medicare phases — MAGI sits between 100% and 400% FPL. ✓

▸ Why this matters

IRMAA Not Triggered

MAGI stays below IRMAA thresholds in all phases — no Medicare surcharges triggered. ✓

▸ Why this matters

Portfolio Survives to End

Portfolio balances remain healthy through all phases. ✓

▸ Why this matters

RMD Compliance

RMD shortfall of $726/mo. IRS penalties (25% of shortfall) apply. Increase 401k withdrawals in the RMD phase immediately.

I'm OK with this ▾ Hide explanation

From age 73 the IRS requires you to withdraw a minimum amount from every Traditional 401k/IRA each year. The check flags any shortfall.
Why it matters: the penalty for missing an RMD is **25% of the shortfall** (10% if corrected within two years). Most people don't know about RMDs until the IRS reminds them.

▶ RMD · 2 min ▶ Roth conversion · 3 min

Income Floor Foundation

Guaranteed income covers 105% of average monthly income — a strong, resilient floor that protects the majority of your income from market risk. ✓

▸ Why this matters

Stress Test Resilient

Drawn-from accounts survive a bear scenario (returns −3%, inflation 5%) to age 80. The plan has meaningful resilience against adverse conditions. (Untouched legacy balances are

Consciously dismiss a check that doesn't apply — with an optional note for your records.

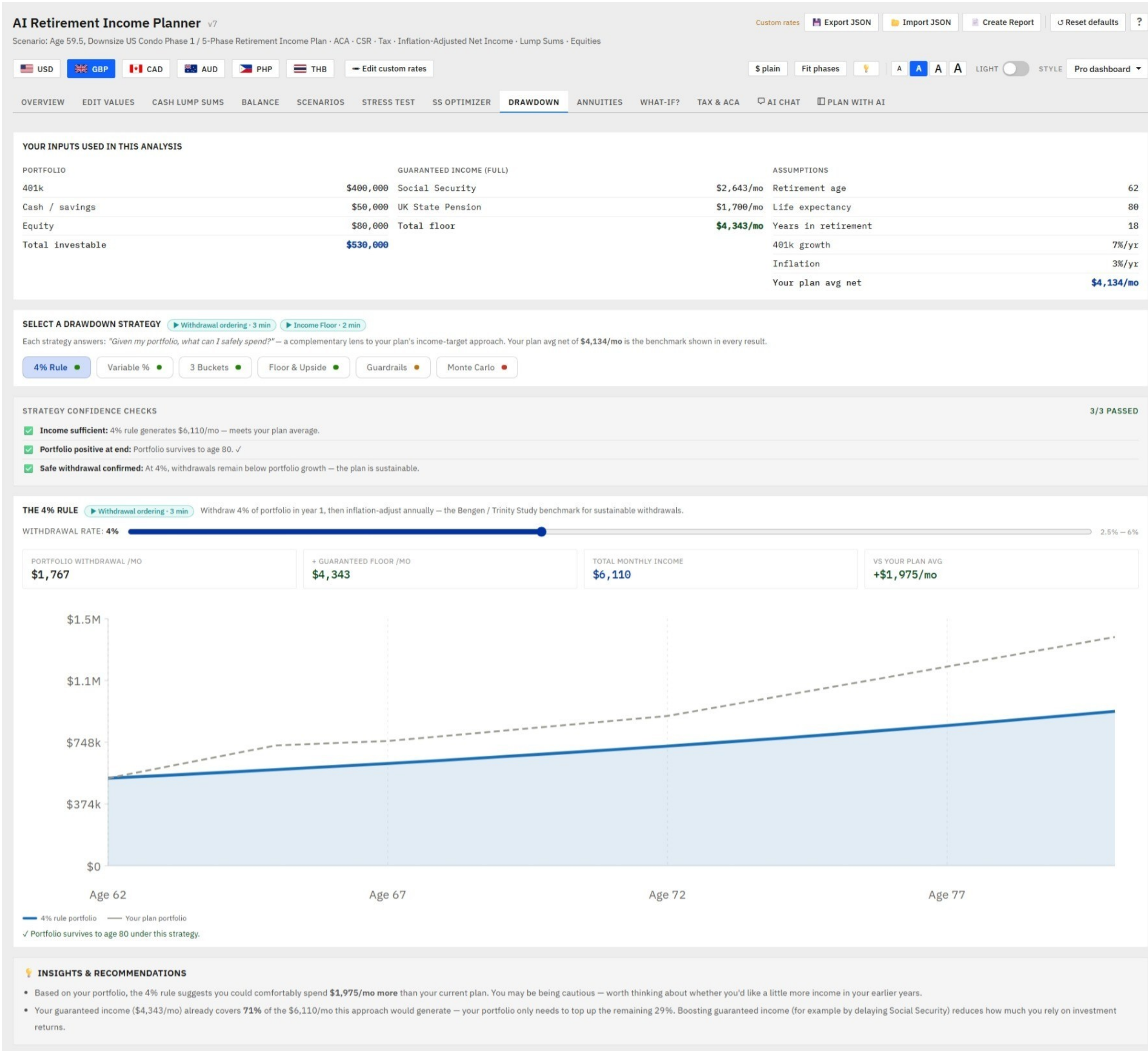
Chapter 13 — The five drawdown strategies

The **Drawdown** tab answers a complementary question: "*Given my portfolio, what do established withdrawal strategies say I can safely spend?*" Five deterministic strategies plus Monte Carlo are available.

- **4% Rule** — withdraw a fixed % of your starting portfolio in year 1, then inflation-adjust. Slider 2.5%–6%.
- **Variable Percentage Withdrawal (VPW)** — an increasing % calibrated to growth and horizon; reaches zero at life expectancy.
- **Three-Bucket** — cash funds years 1–3, Roth/low-risk funds years 4–10, growth funds year 11+.
- **Floor & Upside** — separates guaranteed income from portfolio-dependent income with a coverage table.
- **Guardrails** — starts at 5% with rules to cut/raise spending 10% as markets move.

Plain-language insights

Every strategy includes an Insights & Recommendations section in plain English — e.g. "*Even in the worst case, income would only drop to $5,800/mo — within 15% of your current plan.*"



Five established withdrawal strategies, each explained in plain English.

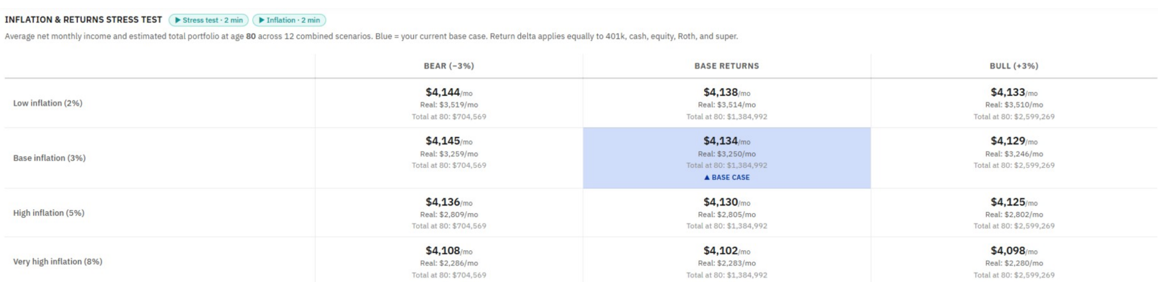
Chapter 14 — Stress Test: what 12 scenarios reveal

The **Stress test** tab runs your plan through 12 combined scenarios: four inflation rates (2%, your base, 5%, 8%) crossed with three return rates (bear −3%, your base, bull +3%). Each cell shows average monthly net income (nominal and real) and projected final-phase portfolio value.

Reading the grid

Your base case is highlighted. Upper-right cells (higher inflation/returns) tend to favour your plan; lower-left cells are the risk scenarios. If your plan looks healthy in the bottom-left corner, your strategy is robust to simultaneous adverse conditions.

> **Nominal vs real**
>
> Big nominal drops in high-inflation cells aren't necessarily bad — check the real number for true purchasing-power impact.



INFLATION & RETURNS STRESS TEST ▶ Stress test · 2 min ▶ Inflation · 2 min

Average net monthly income and estimated total portfolio at age **80** across 12 combined scenarios. Blue = your current base case. Return delta applies equally to 401k, cash, equity, Roth, and super.

	BEAR (−3%)	BASE RETURNS	BULL (+3%)
Low inflation (2%)	**$4,144**/mo Real: $3,519/mo Total at 80: $704,569	**$4,138**/mo Real: $3,514/mo Total at 80: $1,384,992	**$4,133**/mo Real: $3,510/mo Total at 80: $2,599,269
Base inflation (3%)	**$4,145**/mo Real: $3,259/mo Total at 80: $704,569	**$4,134**/mo Real: $3,250/mo Total at 80: $1,384,992 ▲ BASE CASE	**$4,129**/mo Real: $3,246/mo Total at 80: $2,599,269
High inflation (5%)	**$4,136**/mo Real: $2,809/mo Total at 80: $704,569	**$4,130**/mo Real: $2,805/mo Total at 80: $1,384,992	**$4,125**/mo Real: $2,802/mo Total at 80: $2,599,269
Very high inflation (8%)	**$4,108**/mo Real: $2,286/mo Total at 80: $704,569	**$4,102**/mo Real: $2,283/mo Total at 80: $1,384,992	**$4,098**/mo Real: $2,280/mo Total at 80: $2,599,269

Twelve scenarios at once: how your plan holds up across inflation and return combinations.

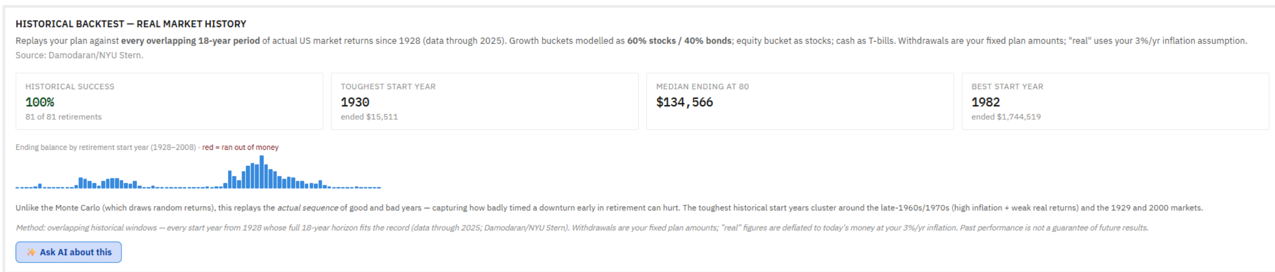
Historical backtest — would your plan have survived real history? NEW

Below the 12-scenario grid, the **Historical backtest** card asks a different question. Monte Carlo (Chapter 15) draws each year's return at *random*; the backtest replays the *actual sequence* of US market returns — every overlapping retirement window since **1928**. That sequence matters: a crash in your first few retirement years does far more damage than the same crash later, and random sampling tends to understate that "sequence-of-returns" risk. It runs automatically — no button to press.

Your **equity** bucket is modelled as 100% stocks, your **cash** bucket as Treasury bills, and your **growth** buckets (401k, Roth, Super) as a stock/bond blend you set with **Backtest growth-bucket stock %** (default 60/40; Chapter 7). Returns come from the Damodaran / NYU Stern public dataset, updated through the most recent complete year. Four tiles summarise the run: **historical success** (how many windows survived, colour-coded green ≥90% / amber 70–89% / red below), **toughest start year**, **median ending** balance, and **best start year**. A strip of bars below shows the ending balance for each start year, with red bars flagging windows that ran out — the late-1960s/1970s, 1929 and 2000 cohorts usually stand out.

Backtest vs. Monte Carlo — use both

Monte Carlo asks "across thousands of *random* futures, how often does my plan work?" The backtest asks "if the next decades look like any we've actually lived through, would I have made it?" A confident plan does well on both. If the backtest is noticeably weaker, sequence risk is your vulnerability — consider a larger early-years cash buffer or a slightly lower early withdrawal. (If your plan horizon is longer than the available history, the card says so rather than showing partial results.) An ✨ **Ask AI about this** button hands the result to the AI.



Replay your plan against every retirement since 1928 — the toughest start years show exactly where sequence risk bites.

Chapter 15 — Monte Carlo simulation

Monte Carlo runs up to 2,000 random-walk simulations (default 500). In each, an annual return is drawn independently for each bucket from a normal distribution with mean equal to your configured rate and standard deviation equal to your configured sigma.

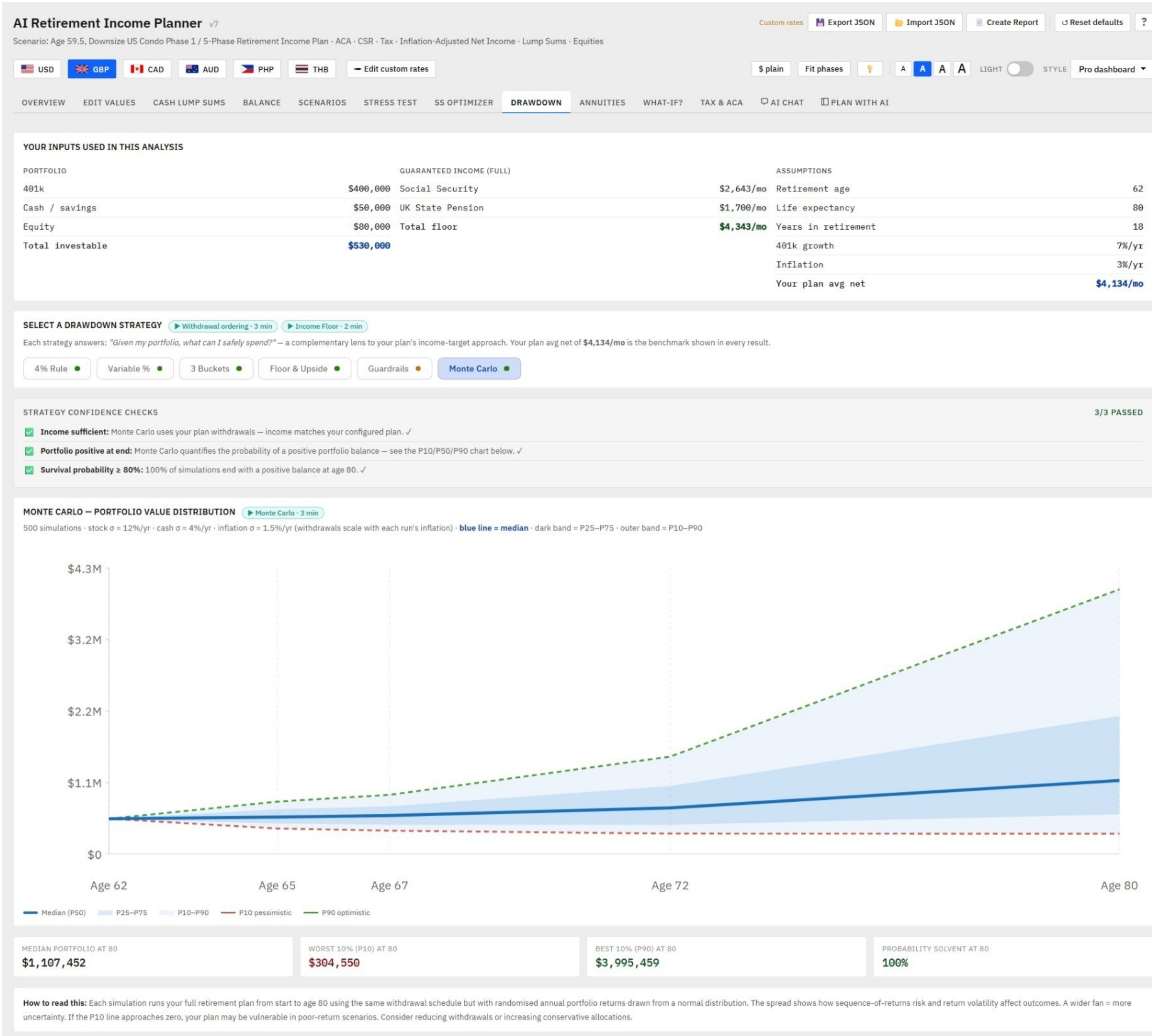
Reading the fan chart

The chart shows percentile bands P10/P25/P50/P75/P90. The headline number is the probability of solvency — the percentage of runs ending with a positive balance at your final phase.

Volatility settings

Each account has its own sigma on the Edit values tab: equity typically 12–18%, 401k/bonds 6–10%, cash 2–4%. The new ⓘ popover on the cash/bond volatility field explains sensible ranges (Chapter 28).

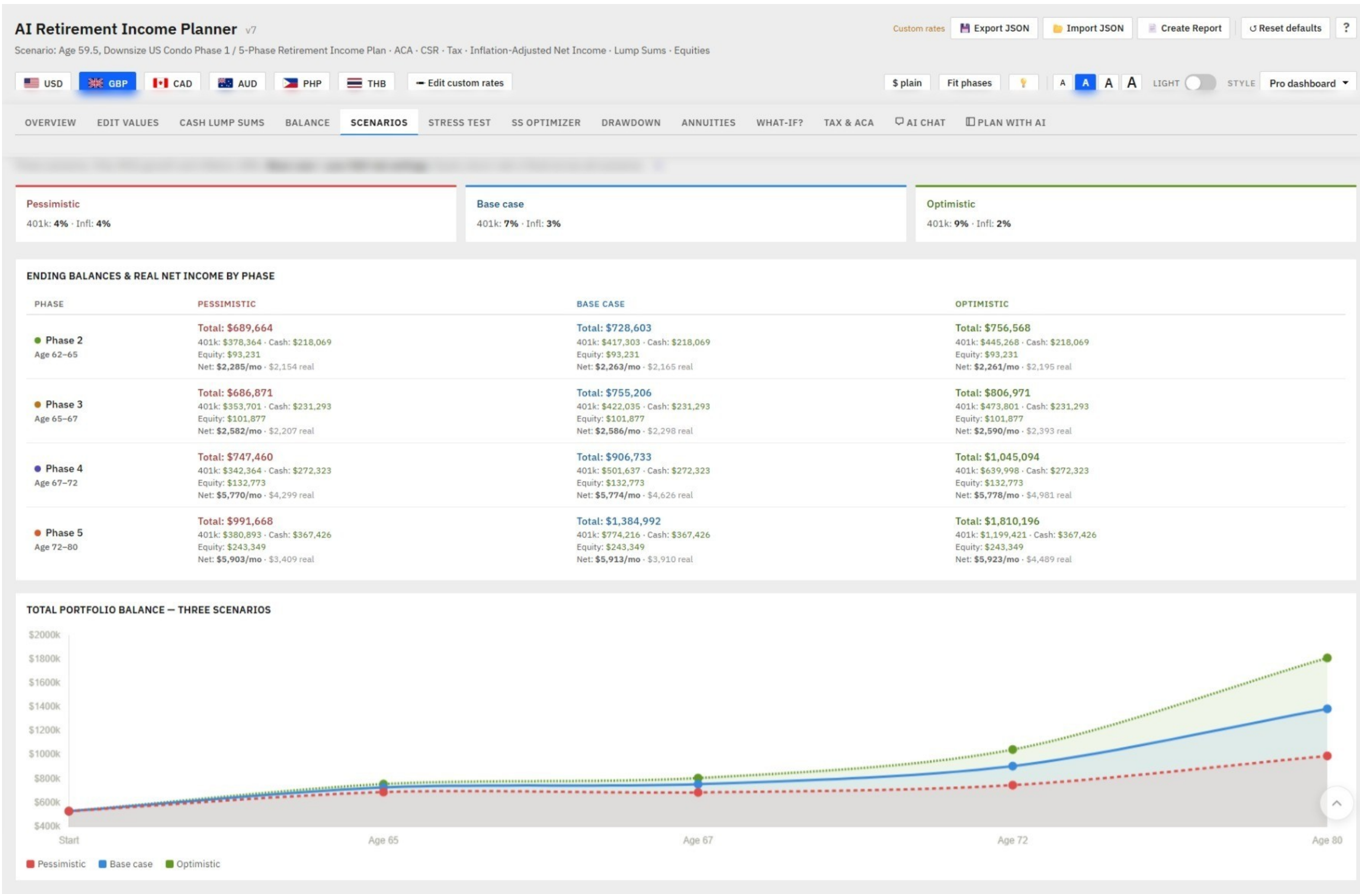
> **What Monte Carlo doesn't do**
>
> It does not predict the future — it quantifies uncertainty given your assumptions. A 92% probability of solvency means 92% of randomly-sampled future paths leave you with a positive balance.



The fan chart shows the range of outcomes; the headline figure is your probability of solvency.

Chapter 16 — Scenarios comparison

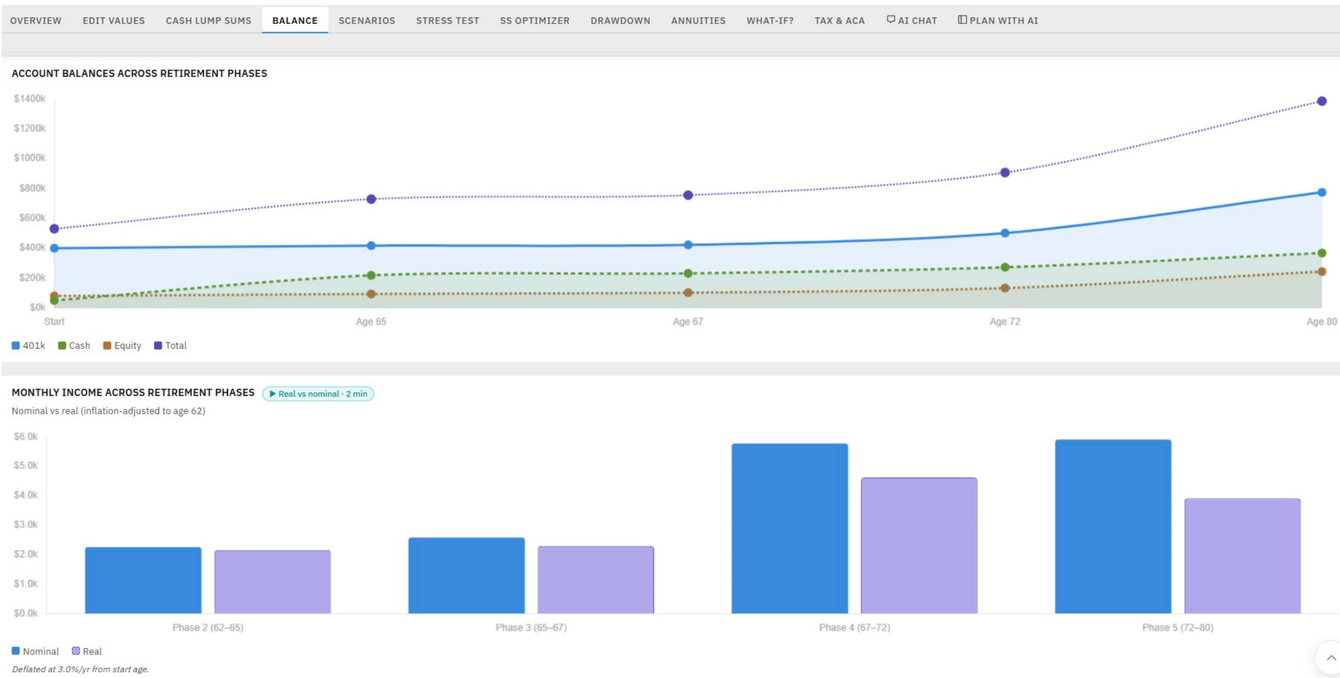
The **Scenarios** tab compares your plan against three pre-defined scenarios — Pessimistic (4% returns, 4% inflation), your Base case, and Optimistic (9% returns, 2% inflation) — side by side. Use it to see how much depends on your return assumption, how much real income you lose if inflation runs hot, and whether the pessimistic–optimistic gap is narrow enough to plan around.



Three return/inflation scenarios side by side.

Chapter 17 — Balance chart

The **Balance** tab visualizes your account balances (401k, Cash, Equity, Roth) over time, month by month. Each account is a coloured line; the X axis is age. Look for lines crossing zero (accounts depleting early), steep drops, rising lines after retirement (growth outpacing withdrawal), and crossover points hinting at rebalancing opportunities.



Watch each account rise and fall across your whole retirement.

Part 4 — AI features

Several features use Claude (Anthropic's AI) to make the planner more accurate, more conversational and more current. All are optional and require an Anthropic API key (Chapter 18). In v7 the same key also unlocks the new **Plan with AI** co-pilot (Chapter 20).

Chapter 18 — Adding your Anthropic API key

The AI features require an Anthropic API key. This is different from a login — you create a developer account at *console.anthropic.com* (free to sign up), generate a key, paste it into the planner once, and you're done. You only pay for usage (typically a few cents per session).

To get a key

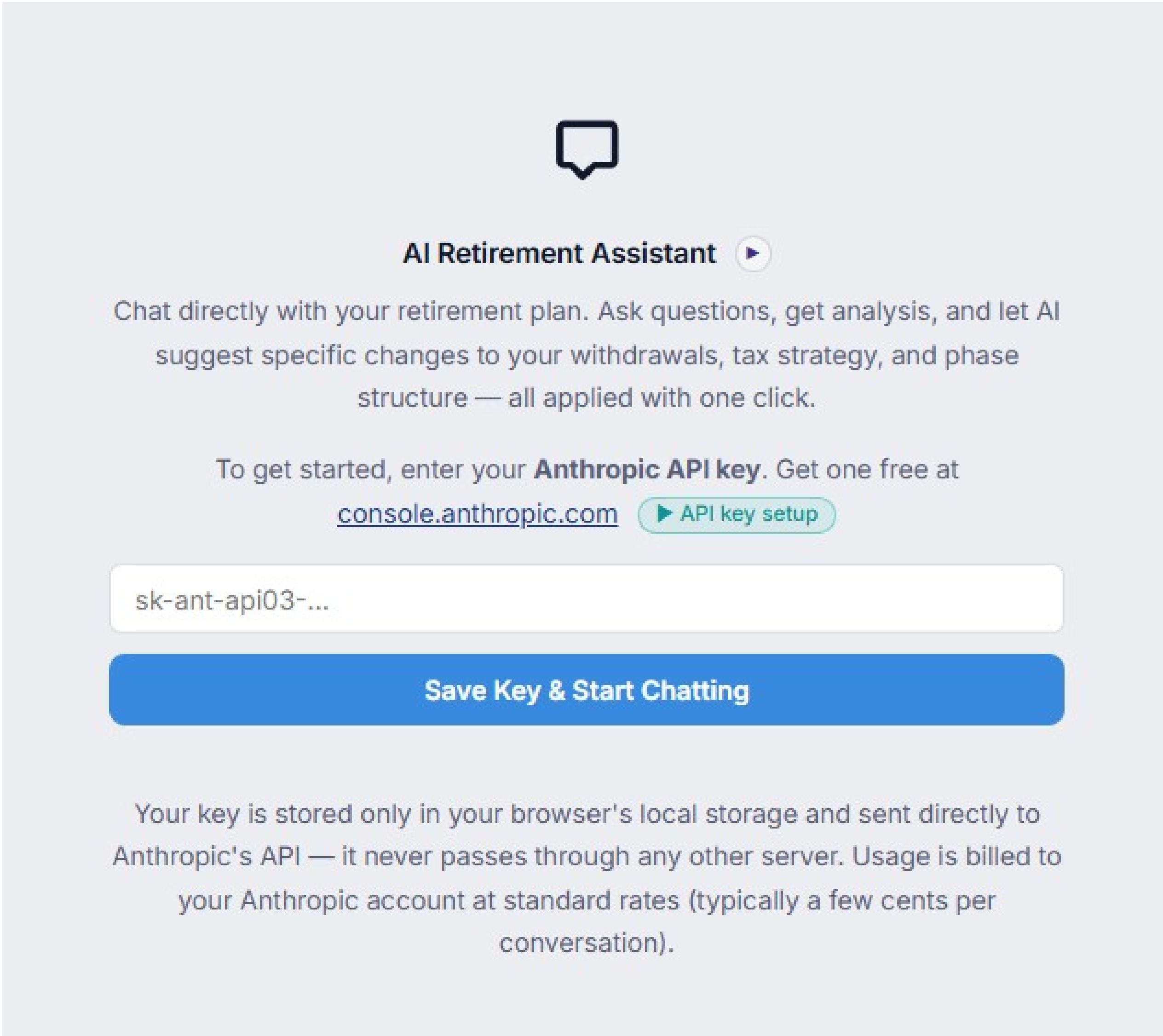
1. Visit *console.anthropic.com* and sign up (or log in).
2. Add a payment method (pay-as-you-go; topping up $5 lasts many months).
3. Go to **API Keys** in the console sidebar.
4. Click **Create Key**, name it, and copy the key (it starts with `sk-ant-`).

To add the key to the planner

1. Open the planner and click the **AI Chat** tab (or open the **Plan with AI** sidebar).
2. Paste your key in the input field, then click **Save key**.
3. The key is stored only in your browser's local storage. It's sent only to Anthropic's API when you use an AI feature — never to any third-party server.

> **One key, every AI feature**
> The same key unlocks AI Chat, the Plan with AI sidebar, the state-tax lookup, Fetch current rates, the Help content audit, and AI commentary in the PDF report. You only enter it once.

> **Keep your key private**
> Treat your API key like a password. Don't paste it into emails, chats or screenshots you share. If a key is ever exposed, rotate it in the Anthropic console.



Paste your key once; it's stored only in your browser and sent only to Anthropic.

Chapter 19 — AI Chat: six starter prompts and how to read proposals

The **AI Chat** tab is the full-screen conversational surface. Every message includes a complete live snapshot of your plan — balances, phases, tax parameters, calculated results and Plan Health checks — so responses are specific to your numbers.

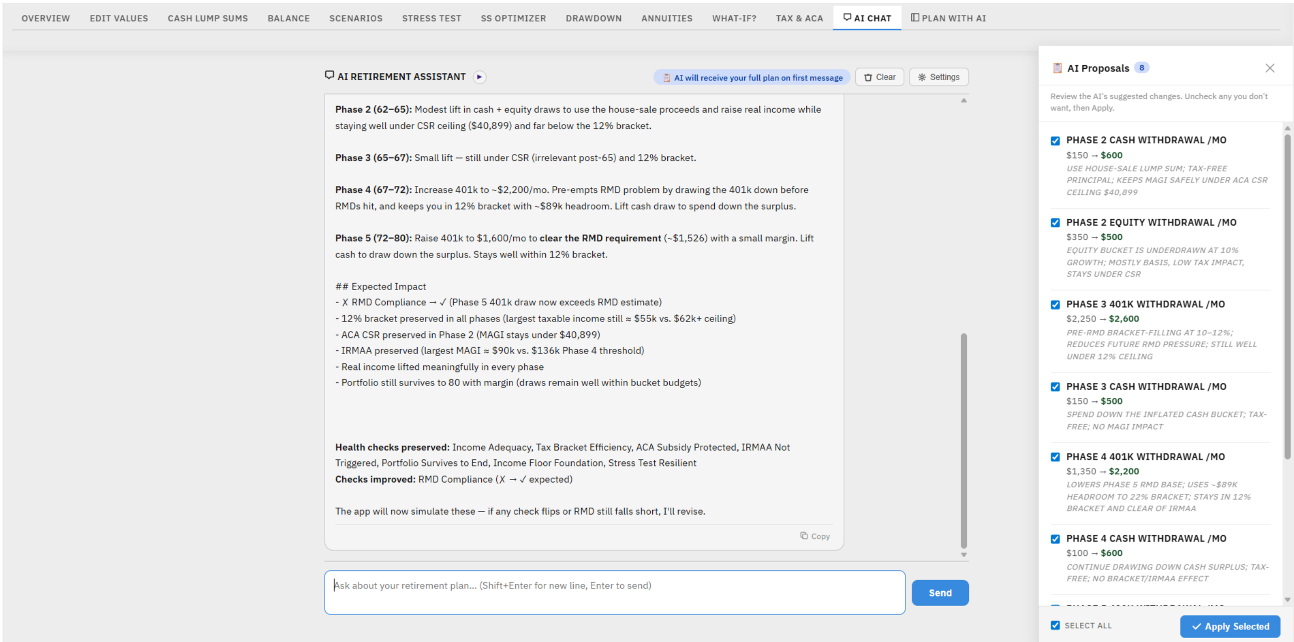
Six conversation-starter pills

- **Summarise & flag risks** — a 60-second overview with the 2–3 biggest risks.
- **Tax bracket check** — per-phase bracket distance and unused headroom.
- **ACA / CSR check** — cliffs and CSR opportunities in pre-Medicare phases.
- **Withdrawal strategy** — suggested adjustments to your account mix.
- **Propose optimizations** — specific numeric changes you can apply with one click.
- **Roth conversion advice** — conversion amounts that fit your bracket headroom.

AI proposals — applying changes directly

When you ask for optimizations, Claude returns a structured proposal (e.g. "*Reduce 401k withdrawal in Phase 2 by $3,000, increase Roth by $3,000 — keeps you below 400% FPL*") with an **Apply** button. AI proposals are constrained by your Plan Health state — Claude won't push you over the ACA cliff or into a higher IRMAA tier unless you explicitly ask.

> **New in v7 — copy any AI response**
> Each AI response now has a **copy** icon at the bottom. Click it to copy a clean, plain-text version of the reply to your clipboard — ready to paste into Word, email or your planning notes.



Ask for optimizations, review the proposed changes, apply with one click — and copy the explanation to keep.

Chapter 20 — Plan with AI: the co-pilot sidebar NEW

The headline feature of v7 is **Plan with AI** — a dockable AI co-pilot that lives in a sidebar so you can keep your dashboard, phase cards or charts fully visible while you ask the AI to change your plan in plain English. Think of it as a planning assistant that sits beside your work, like the AI helpers in modern coding tools.

Opening and docking the sidebar

Click **Plan with AI** at the right-hand end of the tab row. The sidebar slides in from the left; your dashboard shifts over so nothing is hidden. The input box is always visible at the bottom. Click the button again (or the collapse control) to tuck it away. Your open/closed preference is remembered between sessions. On a phone the sidebar overlays the page and closes when you tap outside it.

A separate conversation from the AI Chat tab

The sidebar keeps its *own* conversation history, independent of the AI Chat tab — so a quick "bump SS to 67" in the sidebar won't clutter a longer analysis you're running in the tab. It has its own **clear-chat** button, a **copy** icon on responses, and an ⓘ **info** link to the in-app help for the sidebar.

Action-oriented quick chips

When the sidebar is empty it shows action chips that *change your plan* rather than just answer questions — for example "*rebalance so equity can grow*" or "*optimise my pre-Medicare phases for ACA*". The first, most prominent chip is a getting-started prompt: "*Start here: add your monthly withdrawals in Edit values or directly on the dashboard.*"

Verified proposals — the AI tests its own advice

This is what makes the co-pilot trustworthy. Because the planner's calculation engine runs instantly on your computer, the sidebar **tests every proposal in the simulator before you apply it**:

1. It applies the AI's proposed changes to a private copy of your plan and re-runs the full month-by-month projection.
2. It reads the result and shows a small **outcome badge** on the proposal — green if your plan stays solvent to your plan horizon, amber/red if any account would still run dry.
3. If you asked for a plan that doesn't run out of money and the first attempt still depletes, the co-pilot **automatically tells the AI its suggestion failed and asks it to refine** — up to three attempts — posting a short note each time it retries. You see only verified, workable proposals.

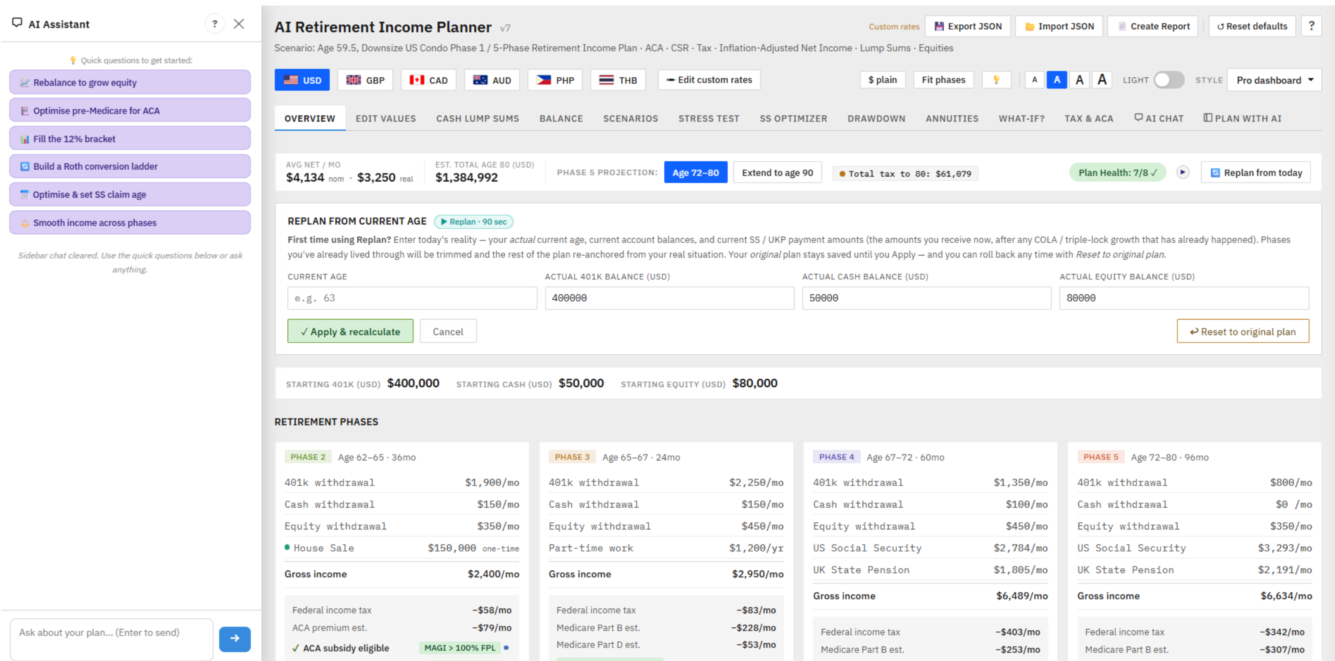
> **Why this matters**
>
> Language models can't reliably do month-by-month compound math in their heads. By checking each proposal against the planner's deterministic engine, the co-pilot won't hand you a confident-sounding suggestion that quietly runs you out of money in Phase 5.

Apply — and one-click Undo

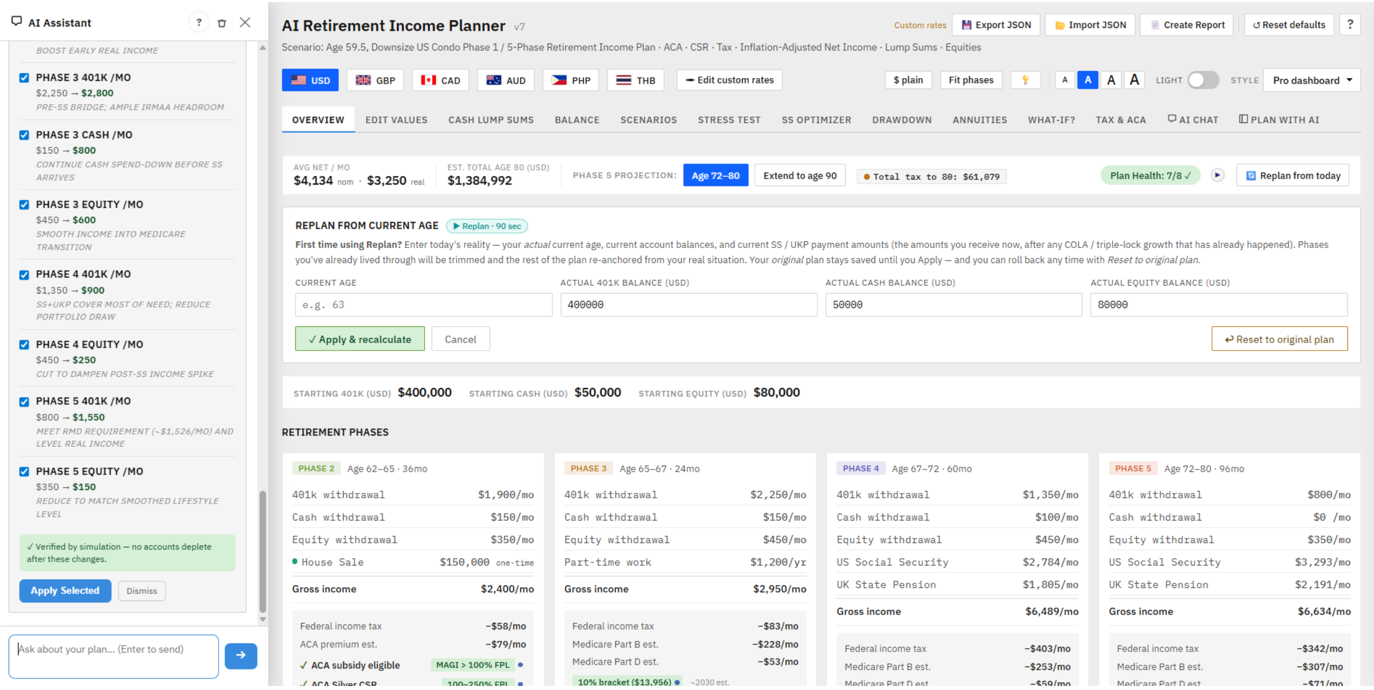
Proposals appear as cards with every change pre-ticked; untick anything you don't want and click **Apply Selected**. The dashboard updates instantly. If you change your mind, a single **Undo** button reverses the entire batch you just applied — so you can experiment freely without manually putting numbers back.

Stays on topic

The co-pilot is focused on retirement planning. Ask it something unrelated and it will politely steer back to your plan.



The Plan with AI co-pilot docks on the left so your dashboard stays in view.



Each proposal is tested in the simulator first — green means your plan still lasts to your plan horizon. Undo reverses the whole batch.

Chapter 21 — Fetch current rates (and state-tax lookup)

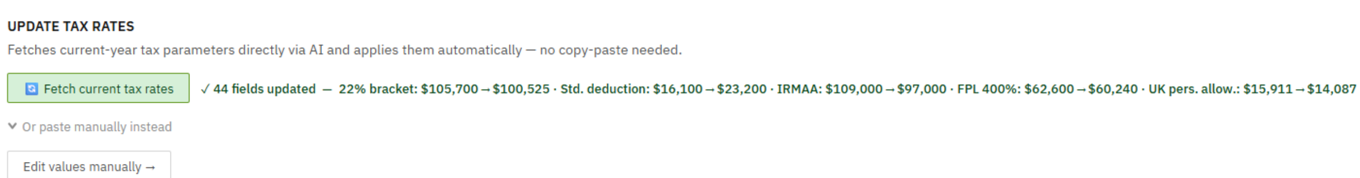
Tax brackets, FPL thresholds, the IRMAA ladder, Medicare premiums and the standard deduction all change every year, and a file you downloaded months ago still holds the figures it shipped with. The **Fetch current rates** button at the bottom of the Edit values tab handles them in one click.

1. On the Edit values tab, scroll to the bottom and find **Fetch current rates**.
2. Click it. The AI looks up current-year values for federal brackets, the poverty levels behind ACA subsidies, the full six-bracket IRMAA ladder, the standard and senior deductions, Medicare Part B and Part D premiums, the RMD age, and UK / Canadian / Australian parameters. Where your AI provider supports it the lookup searches the web, which matters because every one of these figures is reset annually and a model answering from memory will give you last year's. It asks only for the figures your plan can use unless you tick Ask for every country.
3. A review table appears listing every figure it wants to change, what you have now beside what it proposes, and a tick box against each one. Anything that fails a plausibility check - a value outside the range that figure can sensibly take, brackets that do not rise in order, a jump far larger than a year's indexation - arrives already unticked with the reason beside it. 4. Press Apply and only the ticked rows are written — nothing changes until you do.

Without an API key

The same area shows the **Rate Research Prompt** — a paste-ready prompt for any external AI (Claude.ai, ChatGPT, Gemini). Copy it, paste in any AI, copy the response back, and the planner imports the values.

> **New in v7 — state income-tax lookup**
>
> Alongside the federal fetch, the **State** field on the Tax parameters card has its own **Auto-fill rate** button that looks up your state's flat-rate approximation and retirement-income exemptions (see Chapter 7). It's enabled only once you've entered a state.



One click proposes every tax parameter, shows you exactly what would change, and waits for you to accept it.

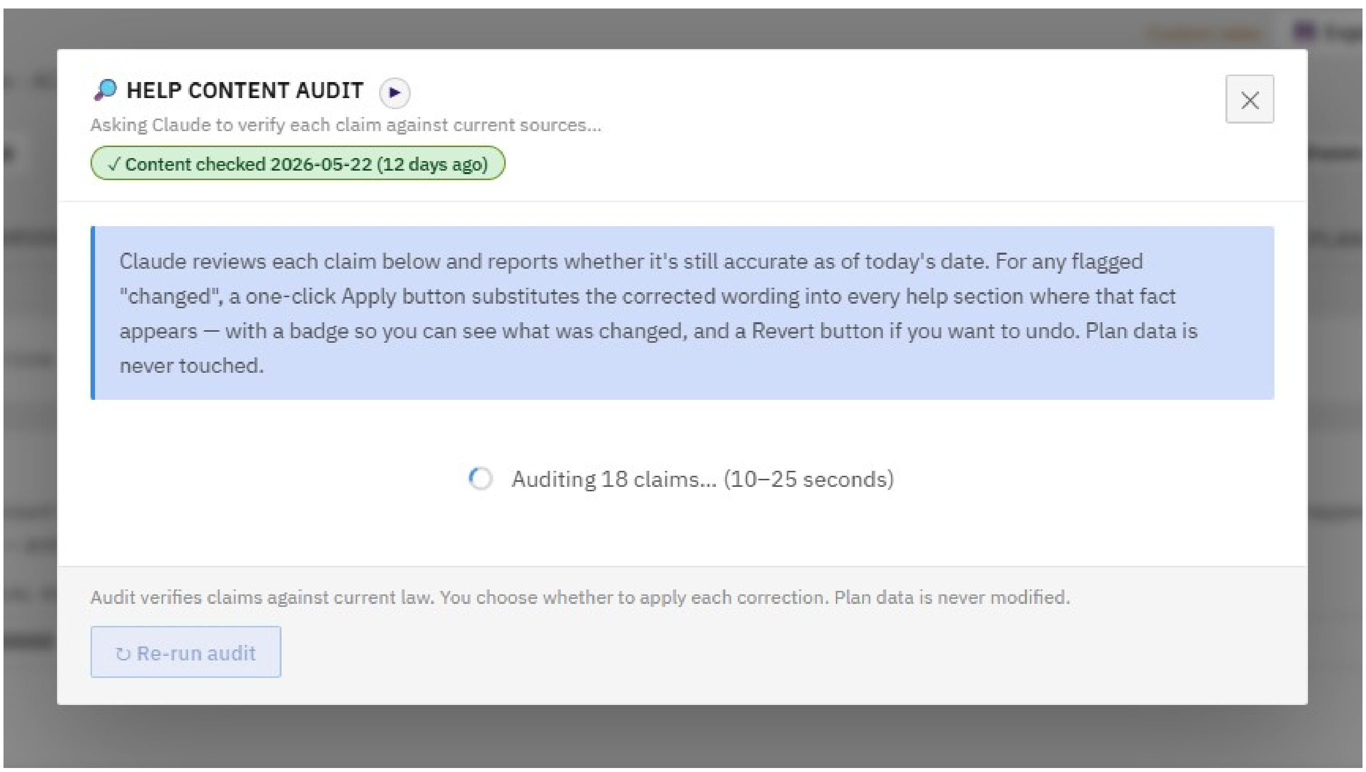
Chapter 22 — Help content audit and one-click corrections

Tax brackets are one form of drift; structural rules and dates (RMD age, ACA cliff status, IRMAA mechanics, treaty articles, recent COLA figures) are another. The **Help content audit** handles those — and can apply corrections to the planner's help prose with a single click.

To run an audit

1. Click **Audit help content** in the planner header.
2. Wait 10–25 seconds while Claude reviews 15 specific factual claims against today's law.
3. Each claim is shown as *current*, *changed*, or *uncertain*.
4. For any *changed* row, click **Apply this update everywhere** — the corrected prose is substituted into the Tax & ACA notes, Concepts Primer, info popovers and glossary, each with a small *AI-corrected* badge.
5. Click **Revert** on any row to restore the original prose.

If all claims come back current, you'll see a green banner and a persistent header badge showing the last audit date. After ~18 months the badge turns amber as a gentle stale reminder. Corrections survive reload and travel with your plan in JSON.



Claude fact-checks 15 claims against today's law — apply corrections everywhere with one click.

Chapter 23 — The PDF report (now with allocation, resilience & charts)

The **Create Report** button produces a clean, multi-page reference report you can keep, email or hand to an advisor. With an API key you can opt in to AI-enhanced commentary that adds short interpretive prose without inventing any numbers. Version 7 substantially expands what the report contains.

What's in the report

- **Cover** — scenario title, generation date, currency, residency, headline metrics, and (optional) a 60-word AI plan summary.
- **Phase-by-phase plan** — one printable card per phase with income sources, tax, healthcare and balances.
- **Top risks & recommended actions** — drawn from your Plan Health failures, with optional AI narratives.

> **Asset-allocation mix** v7 — your portfolio composition at the start and end of the plan, both by account (401k / Cash / Equity / Roth / Super) and by tax treatment (tax-deferred / taxable / tax-free), with a one-line tax-diversification note.

> **Plan-resilience page** v7 — opens with a plain-English **Confidence summary** that combines all three lenses (Plan Health checks passed, the Monte Carlo success rate, and the historical backtest result) and names the main thing to watch — followed by the Monte Carlo detail ("succeeded in N% of simulations to age X" with P10 / median / P90 ending portfolios), the 12-scenario stress-test grid with your base case highlighted, and a historical-backtest summary (how many real retirements since 1928 the plan survived, with the toughest start year).

> **Portfolio charts** v7 — the balance and income charts rendered into the report.

- **Key assumptions** and a **disclaimer** (noting that Monte Carlo / stress results are scenario analyses, not guarantees).

> **Fixed in v7 — no more blank pages**
> Earlier versions could emit blank or clipped pages when printing the report. v7 rebuilds the report to flow naturally across pages, so the printed PDF matches the preview.

> **What the AI adds — and doesn't**
> AI commentary is interpretive prose only. It never generates new dollar figures (all numbers come deterministically from your plan), and the entire AI text appears in an inline-editable preview before going to PDF. Typical cost: about $0.10 per export.

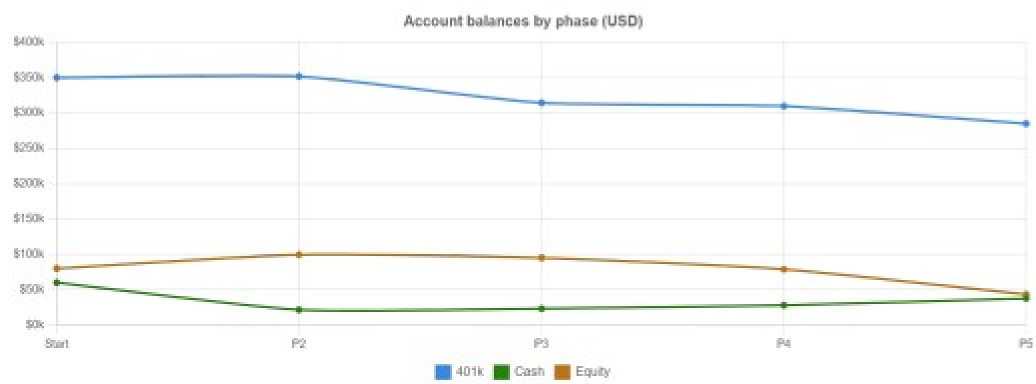
Balance trajectory

Projected account balances at the end of each phase, based on your withdrawal strategy and growth assumptions. A zero balance in an income-producing phase is flagged — see notes below.

PHASE	401K	CASH	EQUITY	TOTAL
Starting (age 62)	$350,000	$60,000	$80,000	$490,000
Phase 2 (ages 62–65)	$351,664	$21,484	$99,498	$472,646
Phase 3 (ages 65–67)	$314,461	$23,155	$94,980	$432,596
Phase 4 (ages 67–72)	$309,762	$27,922	$78,834	$416,518
Phase 5 **(ages 72–80)**	**$285,013**	**$37,673**	**$43,305**	**$365,990**

Portfolio charts

Visual summary of how your account balances and net income evolve across the plan.



Plan resilience

How the plan holds up under uncertainty — random return sequences (Monte Carlo) and combined inflation / return stress scenarios. These are scenario analyses, not guarantees.

Monte Carlo simulation

Across **500** randomized simulations to age 80, your plan ended solvent (money still remaining) in **100%** of runs — a strong result.

PROBABILITY SOLVENT **100%**	WORST 10% (P10) AT 80 **$39,710**
MEDIAN (P50) AT 80 **$394,292**	BEST 10% (P90) AT 80 **$2,130,581**

Each simulation draws annual returns for each account from a normal distribution around your assumed rates. Because the simulation is random, this percentage may vary by a point or two between reports.

v7's report adds allocation, resilience and charts — the kind of pages a professional advisor report includes.

Part 5 — Specialized tools

Several tabs go deeper than the core planner: the Annuity Calculator, the What-if Explorer, the SS Optimizer, and the Roth Conversion Optimizer integration.

Chapter 24 — Annuity Calculator

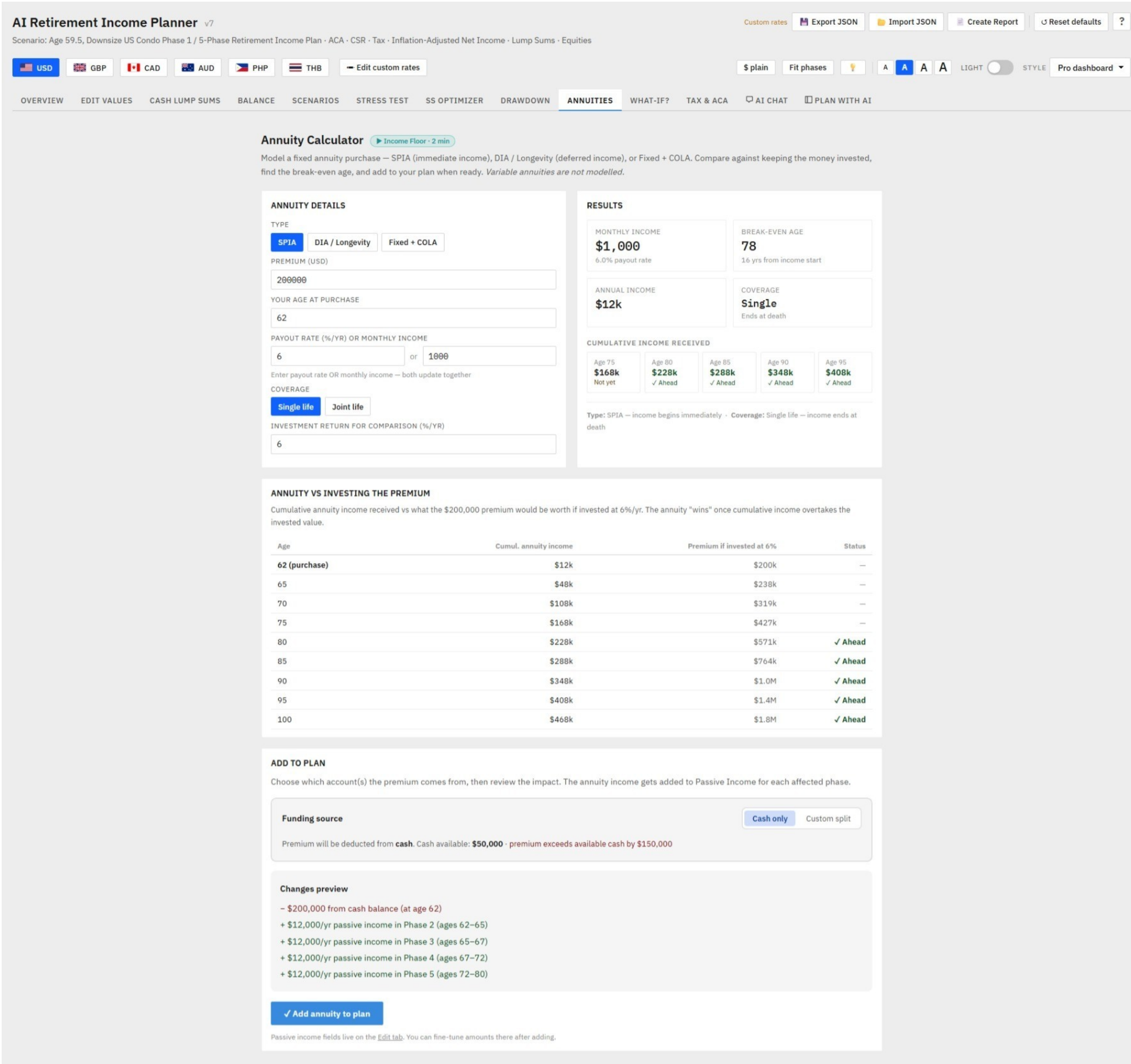
The **Annuities** tab models fixed annuity purchases — SPIA (immediate income), DIA / Longevity (deferred income), or Fixed + COLA (income that grows annually) — and compares them against keeping the money invested. Variable annuities are not modeled.

Setting up an annuity

1. Click the **Annuities** tab.
2. Choose annuity type: SPIA, DIA / Longevity, or Fixed + COLA.
3. Enter premium, purchase age, payout rate (or monthly income — both update together).
4. For DIA set the income-start age (typically 80 or 85); for Fixed + COLA set the COLA rider rate.
5. Choose coverage: Single life or Joint life.
6. Review monthly income, payout rate, break-even age, and the year-by-year comparison versus investing.

Adding to (and removing from) your plan

The **Add to Plan** section deducts the premium from your funding source(s) — cash only, or a custom split across cash, 401k, equity, Roth and any lump sum — and adds annuity income to passive income for each phase it's active in. Every lump sum in your plan appears as a fundable source; allocate part of a lump and the annuity sizes down to match. A **Remove annuity from plan** button reverses every change cleanly.



Model SPIA, DIA or COLA annuities and see the break-even age versus staying invested.

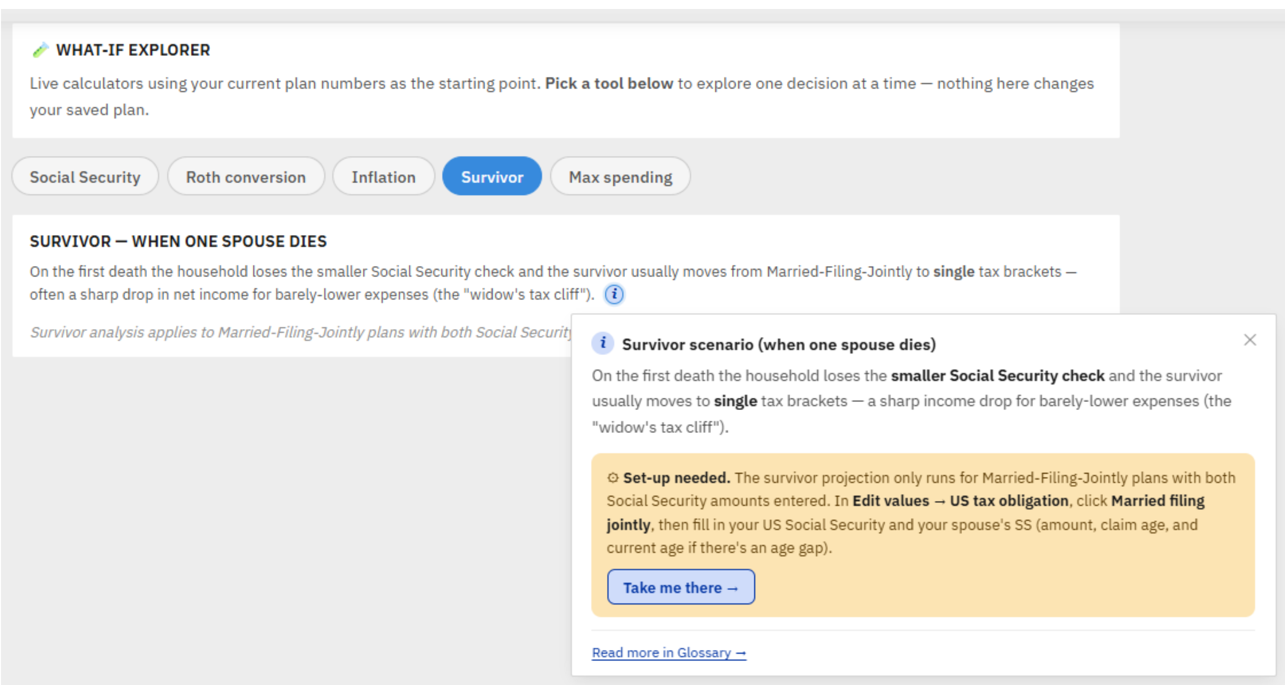
Chapter 25 — What-if Explorer (and the survivor scenario)

The **What-if?** tab is a sandbox for stress-testing specific decisions without modifying your actual plan. Several live tools let you explore the most common "what should I do?" questions — three sliders plus the new Maximum-sustainable-spending finder (below):

- **Social Security claim age (62→70)** — monthly benefit, cumulative payments by 85 and 90, and break-even age.
- **Roth conversion size (per phase)** — tax cost, MAGI shift, ACA / IRMAA status change, net income impact.
- **Inflation rate** — how a 1% change shifts real income at the end of your plan.

Move a slider and the right-hand panel updates instantly — no Calculate button, no waiting. Nothing here writes back to your saved plan.

> **New in v7 — survivor-scenario setup guidance**
> The survivor scenario (what happens when one spouse outlives the other) gives accurate tax results only once your filing status and US tax obligation are set up. Its information popup now explains this and, when that section isn't filled in yet, shows a **Take me there →** link that jumps straight to the US tax-obligation section. The same guidance is mirrored in the Help section and Glossary.

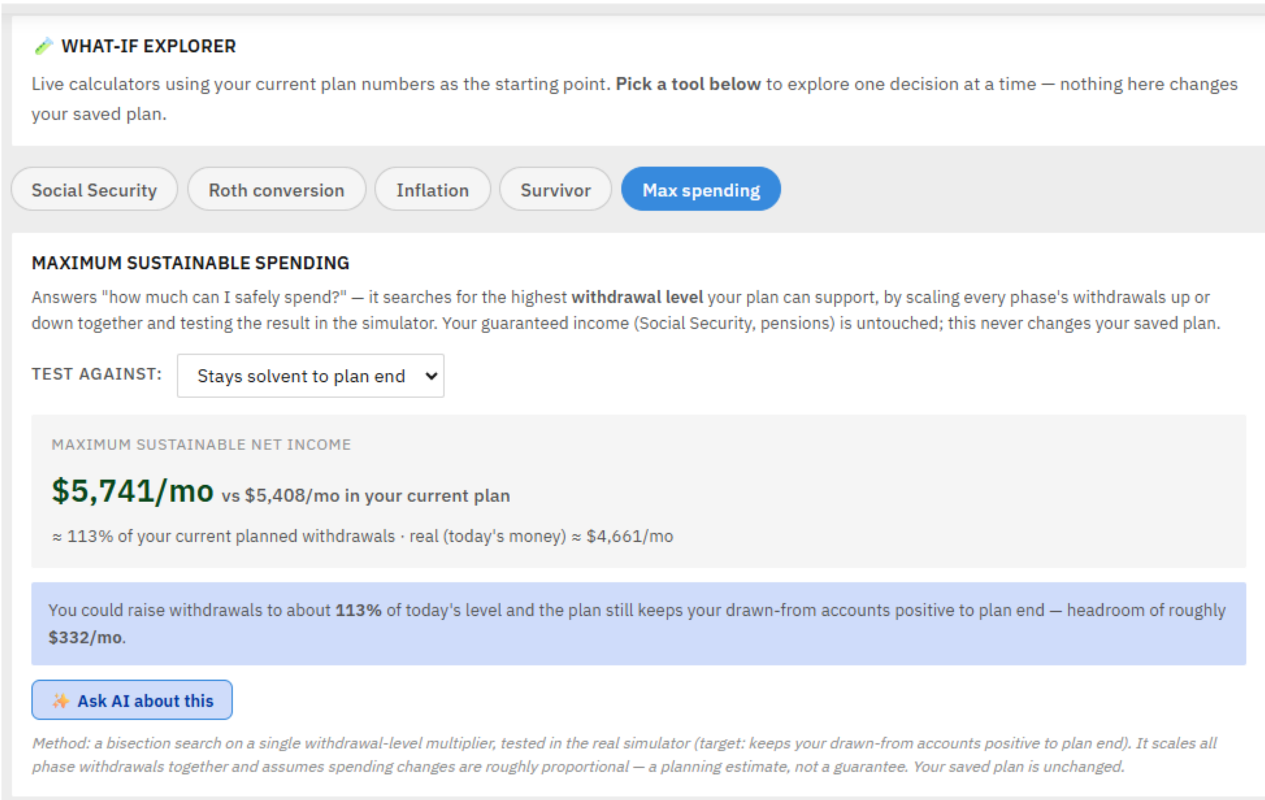


The survivor scenario now guides you to set up your US tax obligation first — with a one-click jump to that section.

Maximum sustainable spending NEW

This card answers the question behind every retirement plan — *"how much can I safely spend?"* It searches for the highest **withdrawal level** your plan can support by scaling every phase's withdrawals up or down together and testing each level in the real simulator. Your guaranteed income (Social Security, pensions) is left untouched, and — like the rest of this tab — it never changes your saved plan. Pick what "safe" means with the **Test against** selector: *Stays solvent to plan end* (the default, instant), *Monte Carlo success*, or *Survives market history*. For the last two you can set the success **target %** (defaults: 85% for Monte Carlo, 90% for history) and click **Calculate**. The result shows your maximum sustainable net income versus your current plan average — so you can see at a glance whether you have room to spend more, or by how much to trim. An ✨ **Ask AI about this** button hands the figure to the AI for next steps.

> **It's an estimate, not a solver**
>
> The tool scales all withdrawals proportionally and treats the pass/fail point as a single boundary — a fast, transparent planning estimate, not a multi-variable optimiser. Use it to size the ballpark, then fine-tune individual phases on the Edit values tab.



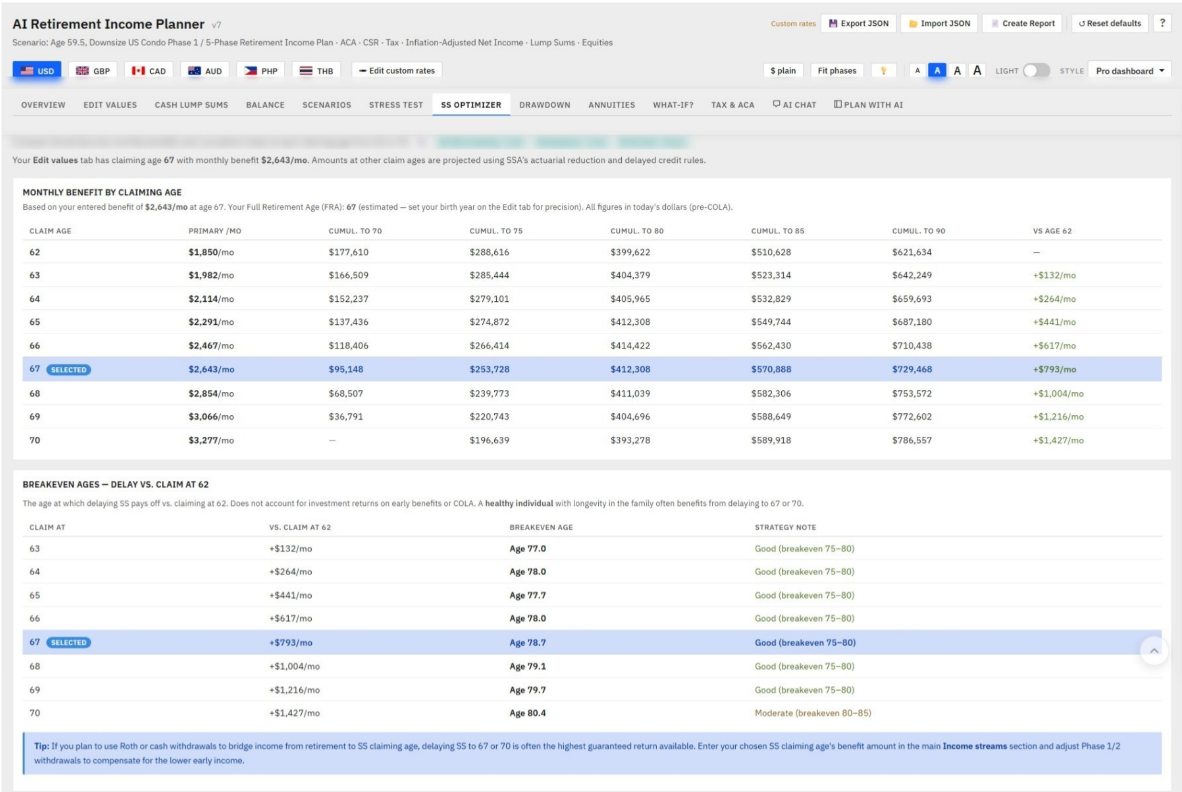
How much can you safely spend? A transparent search for your sustainable withdrawal level — solvent, Monte Carlo, or against real history.

Chapter 26 — Social Security Optimizer

The **SS Optimizer** tab evaluates starting Social Security at each eligible age from 62 to 70. For each starting age it shows cumulative lifetime benefits at three longevity targets — 80, 85 and 90 — and highlights the strategy that produces the most lifetime income at each.

Key insights

- Most people who reach 65 will live past 85 — which favours claiming later.
- Claiming at 62 only "wins" if you live to roughly the early 80s or sooner.
- Delaying to 70 maximises monthly income for life but loses ~8 years of payments; break-even is typically late 70s to early 80s.
- Health, family longevity, and whether you need the money before 70 matter more than the spreadsheet number.



Cumulative benefits at every claiming age, scored against three longevity targets.

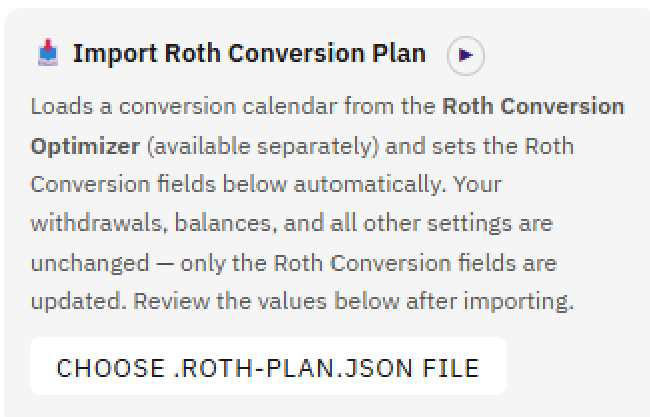
Chapter 27 — Roth Conversion Optimizer integration

Roth conversions are one of the highest-leverage tax decisions in retirement. The planner has its own per-phase Roth conversion field, but designing a multi-year schedule that fills bracket headroom while staying below ACA / IRMAA thresholds is a problem in its own right. The companion **Roth Conversion Optimizer** (sold separately, or as a bundle) is purpose-built for it.

To import a conversion plan

1. Design your strategy in the Roth Optimizer and click **Export plan** to save a `.roth-plan.json` file.
2. In the Retirement Planner, go to the Edit values tab.
3. Find the **Roth Conversion Plan Import** section and click **Choose .roth-plan.json file**.
4. The planner maps each year to the right phase and sets the per-phase Roth conversion fields automatically.

Only the Roth conversion fields are updated; everything else stays as it was. Then open the **Plan with AI** sidebar and ask it to stress-test the imported conversions against your ACA cliff and IRMAA exposure — it sees the new conversions in its plan context.



Import an optimized conversion calendar and the planner fills your per-phase Roth fields automatically.

Part 6 — Learn as you plan

The planner is also a teaching tool. Several overlapping systems explain retirement concepts at the moment they matter, using your actual numbers wherever possible.

Chapter 28 — ⓘ icons: click any field you don't understand

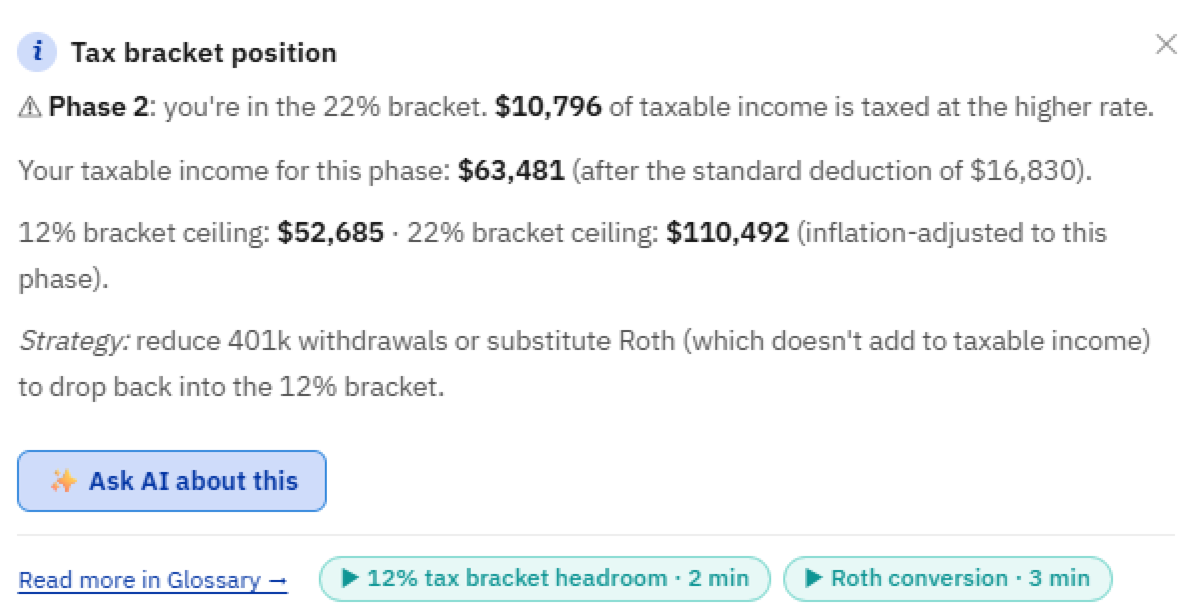
More than two dozen fields show a small ⓘ icon next to their label. Click it for a ~60-word plain-English explanation of what the field is, why it matters, and a common misconception. Most popovers include a **Read more in Glossary** link.

Examples

- ⓘ next to *Roth conversion* explains conversions and how they raise MAGI. ⓘ
- next to *IRMAA threshold* explains the 2-year lookback and the cliff nature.
- ⓘ next to *ACA 400% FPL cliff* explains the rule and the Roth-below-the-cliff strategy.
- ⓘ next to *Equity gain %* explains the embedded-gain concept.

New in v7 — more (and more precise) popovers

- A **getting-started** popover that points new users to enter their monthly withdrawals first.
- A **distinct 22%-bracket** explainer — previously the 12% and 22% bracket icons opened the same text; now each explains its own ceiling.
- A **dedicated SS-claiming-age** popover — the SS claiming-age icon no longer opens the same FRA text as the birth-year field; each now has its own focused explanation.
- A **cash/bond volatility (σ)** popover explaining sensible Monte Carlo ranges for lower-risk buckets.



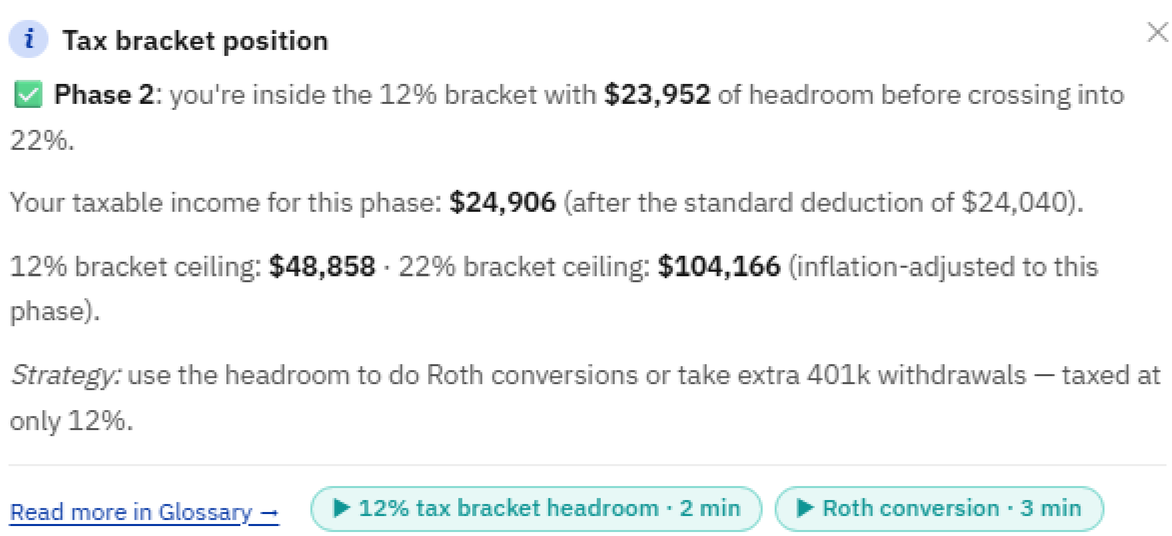
Click any ⓘ for a plain-English explanation — now with separate popovers for the 22% bracket and SS claiming age.

Chapter 29 — Clickable phase-card badges

Every coloured badge on a phase card is clickable. Unlike the ⓘ icons (which explain a concept generically), badge explanations use the actual numbers from that phase:

- *"You're $4,200 below the 12% ceiling. You could withdraw an extra $4,200 from your 401k or convert that much to Roth without bumping into 22%."*
- *"Your phase MAGI is $58,000 — $2,200 above the 400% FPL cliff at $55,800. Either reduce 401k withdrawal by $2,200 or substitute equivalent Roth."*
- *"Your 401k withdrawal of $15,000 is $4,300 below your estimated RMD of $19,300. Increase to at least the RMD level to avoid the 25% penalty."*

Badges are designed to convert insight into action — where appropriate, the explanation includes a specific dollar amount and direction.



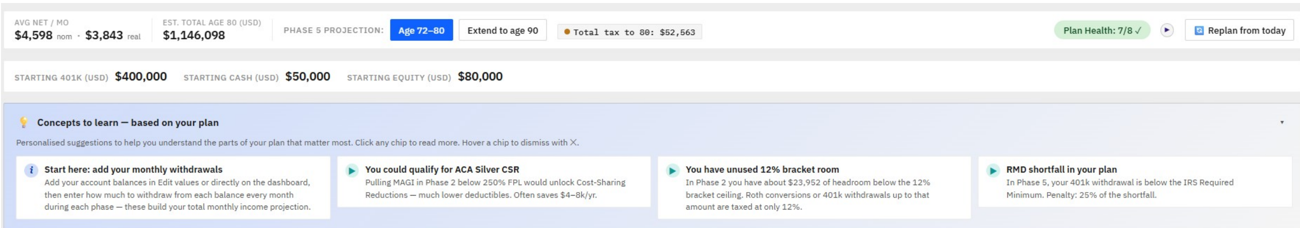
Badges explain your situation in your own numbers — with a specific action to take.

Chapter 30 — Concepts-to-Learn panel

On the Overview tab, a collapsible card titled *Concepts to learn* surfaces 3–4 concepts most relevant to your current plan. The selection is driven by 10+ rules that look at your specific numbers — for example:

- You have a Roth balance but no scheduled conversions → a *Roth conversions in low-bracket years* card.
- You're within 15% of the ACA cliff in a pre-Medicare phase → a *Managing the ACA cliff with Roth* card.
- Your plan ends past 73 with a large 401k and no RMD-aware withdrawals → an *RMD planning before 73* card.
- Your spouse age gap is 3+ years → a *Spouse claim-age timing* card.
- Your later-phase withdrawals are as high as (or higher than) your early ones → a *Model the retirement spending smile* card (see the Glossary entry in Chapter 31).

The card collapses to a thin status bar when you want to focus on numbers, and your choice persists. Each tile opens the relevant ⓘ explanation or glossary entry.



Personalised learning prompts, chosen from your own plan's situation.

Chapter 31 — Glossary and the 9-step Feature Tour

Searchable A–Z Glossary

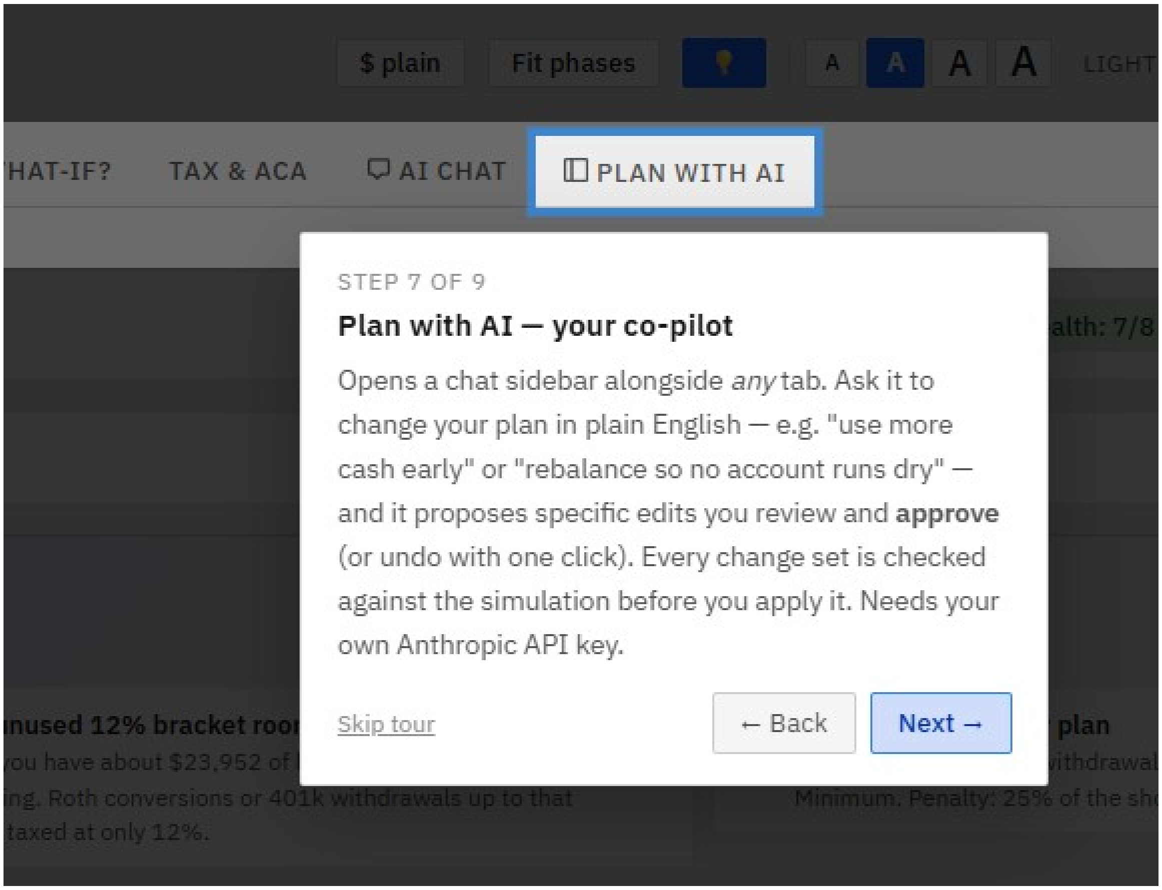
The Glossary tab in the Help modal contains 20+ entries — every retirement term used in the planner, defined with typical values, common misconceptions and related concepts. Type to filter; entries match on term, synonym and body text.

> **New in v7 — the retirement spending smile (go-go / slow-go / no-go)**
>
> Studies of actual retiree spending find it rarely stays flat — it tends to trace a **smile**: the active **go-go** years (travel, hobbies, helping family) cost more; the **slow-go** years settle down; the late **no-go** years can rise again on health and care costs. **How to model it here:** set higher withdrawals in earlier phases and lower ones in the middle, then consider a bump in Phase 5 — or a one-time *money-out* expense (Chapter 8) — for late-life care. Because the planner uses fixed *nominal* withdrawals, even keeping them level lets your *real* spending drift down; shaping the curve deliberately makes the plan match how you'll actually spend. Reachable from the Concepts-to-learn panel (Chapter 30) and this Glossary.

Guided Feature Tour

From the Help modal, click **Take the Feature Tour** for a guided overlay that highlights each major area with a short caption.

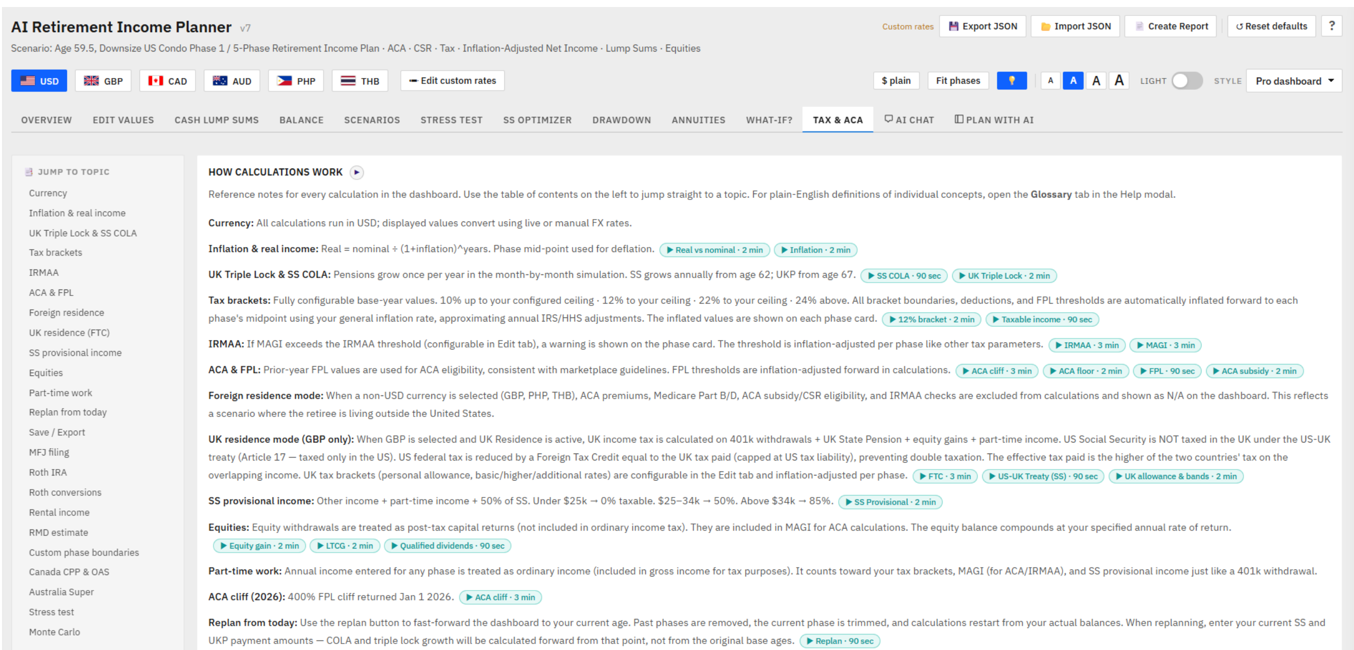
> **New in v7 — expanded to 9 stops**
>
> The tour now covers nine areas, adding stops for the new **Plan with AI** co-pilot and the **Create Report** button alongside the original seven (tab bar, Plan Health badge, Replan button, summary cards, Concepts to learn, phase cards, Help). The tour correctly lifts the sticky header out of the way so highlighted targets are never hidden behind the pinned bars.



The 9-step tour now introduces the Plan with AI co-pilot and the Create Report button.

Chapter 32 — Tax & ACA notes tab

The **Tax & ACA** tab is the technical reference for every calculation in the planner — tax bracket mechanics, IRMAA, SS provisional income, ACA cliff rules, UK residence and the Foreign Tax Credit, equity treatment, replan behaviour and more. If a number on the dashboard surprises you, this is the first place to look. A sticky table of contents on the left lists 23 sections with jump-links; mobile collapses it to a single column. Sentences carry small *AI-corrected* badges where you've applied audit corrections (Chapter 22).



The technical reference behind every number, with a jump-link table of contents.

Part 7 — Settings and maintenance

The last part covers the planner's personalization options, data export/import, and the long-term workflow for keeping your planner current as tax law evolves.

Chapter 33 — Text size, dark mode, themes & the polished header

Text size & dark mode

Buttons in the toolbar (A·A·A·A) set the planner's text size; the moon/sun button toggles light/dark. Both persist across sessions. Dark mode reduces eye strain in low light; light mode prints better.

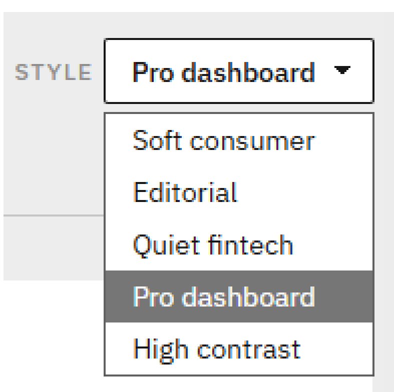
Visual styles (themes)

The **Style** picker offers five visual styles you can switch between at any time:

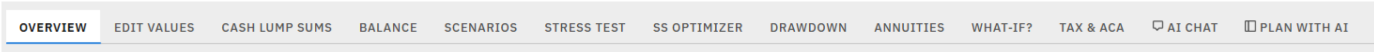
- **Soft consumer** (default) — friendly, rounded.
- **Editorial** — magazine-like.
- **Quiet fintech** — restrained and data-forward.
- **Pro dashboard** v7 — denser, tool-like.
- **High contrast** v7 — an accessibility theme for maximum legibility.

New in v7 — a calmer, clearer layout

- **Workflow tab order:** tabs now read left-to-right as a workflow — inputs (Overview, Edit values, Cash lump sums), then visualize/sensitivity (Balance, Scenarios, Stress test), then strategy (SS Optimizer, Drawdown, Annuities, What-if?), then reference (Tax & ACA), then AI Chat and the Plan with AI toggle.
- **Polished sticky header:** as you scroll, the tab bar lifts with a soft shadow and a frosted background, and a slim marker slides under the active tab. (Motion is automatically disabled if your system prefers reduced motion.)
- **Grouped Overview toolbars:** the controls along the top are grouped by purpose with thin dividers — dashboard-view controls separated from appearance controls, and data/output actions separated from Reset and Help.



Five visual styles — including the new Pro dashboard and High contrast themes.



Grouped toolbars and a polished sticky header keep the controls calm and oriented.

Chapter 34 — Saving, exporting, importing

Auto-save

Every change is automatically saved to your browser's local storage. Re-open in the same browser and your plan is still there — nothing to remember to click.

Export & import plan as JSON

Click **Export JSON** to download a `.json` file containing your complete plan — every balance, withdrawal, tax parameter, lump sum, acknowledged check, AI correction and annuity. Click **Import JSON** to load a previously-exported file (it replaces what's in your browser). Useful for backups, switching scenarios, or moving between machines.

> **Privacy reminder**
> Your plan never leaves your computer unless you explicitly export it. Auto-save writes only to your browser's local storage — not to any cloud service.

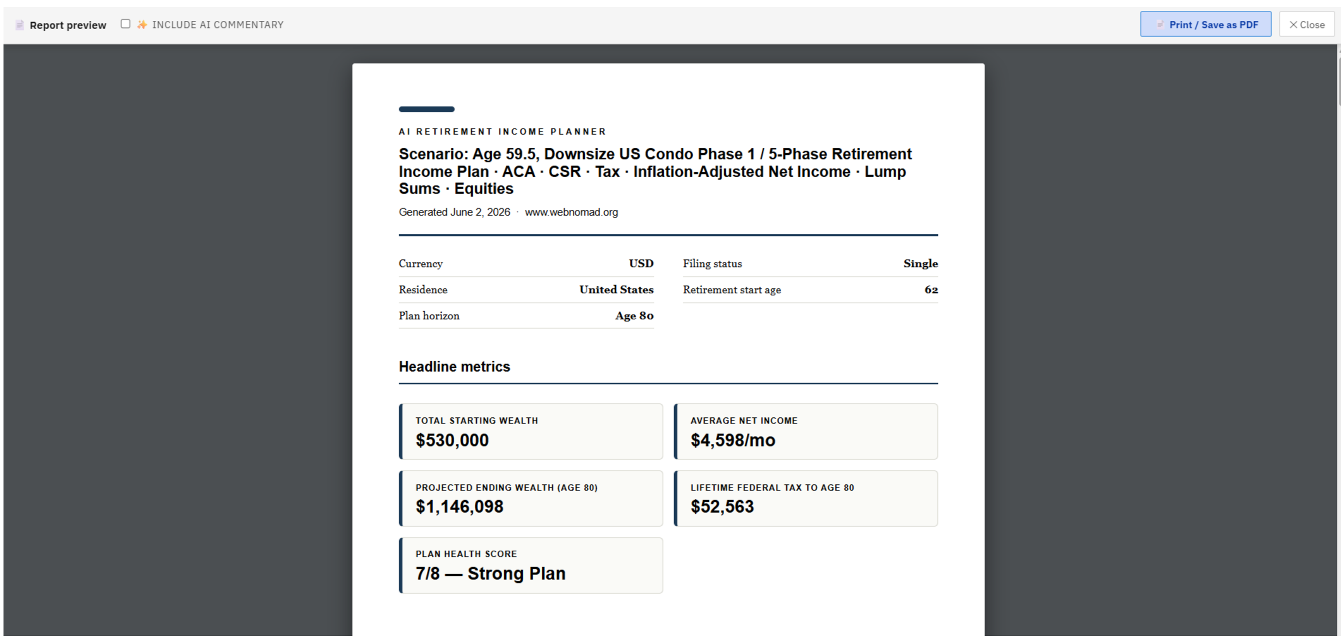
Export a complete JSON backup, or import one to restore or share a plan.

Chapter 35 — PDF export

Create a clean, multi-page report with the **Create Report** button (grouped with Export/Import on the Overview title row).

1. Click **Create Report**.
2. If you have an API key, choose whether to include AI-enhanced commentary (or set a remembered preference).
3. Review the preview — AI prose appears in editable blocks you can adjust or skip.
4. Print to PDF (Save as PDF) and choose where to keep the file.

See Chapter 23 for the full contents — including the new asset-allocation, plan-resilience and portfolio-charts pages, and the blank-page fix.



Choose AI commentary or rule-based only — then review and edit before saving the PDF.

Chapter 36 — Keeping your planner current (the long-term workflow)

Most planners go stale the moment tax law moves. This one has four built-in mechanisms. Use them as a quarterly or annual maintenance routine.

The four mechanisms

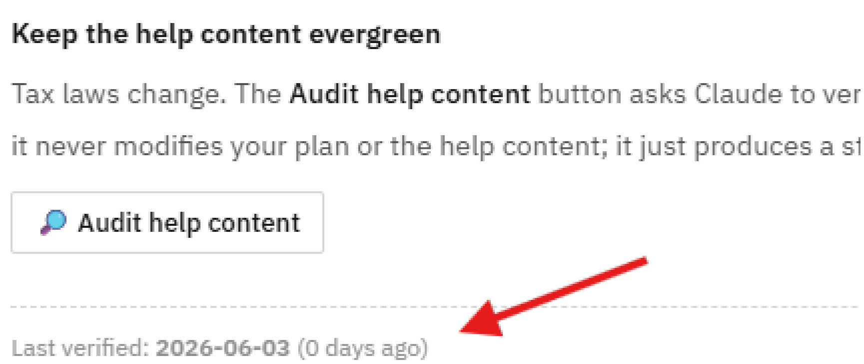
Tier 1 — Auto-updating popovers. When you run **Fetch current rates** (Chapter 21), five info popovers automatically reflect the new numbers — IRMAA threshold, ACA cliff, ACA CSR band, 12% bracket, standard deduction.

Tier 2 — AI audit of structural claims. Click **Audit help content** (Chapter 22) to verify 15 factual claims against today's law. Read-only by default.

Tier 3 — Verified watermark. Every Help pane shows when the help text was last audited (or "never audited — content baseline"). After 18 months the watermark turns amber.

Tier 4 — One-click prose corrections. When the audit finds a changed claim, click **Apply this update everywhere** to substitute the corrected prose into every place that fact appears, with an *AI-corrected* badge. Click **Revert** to undo.

> **Recommended quarterly routine**
>
> 1) Fetch current rates and review the diff. 2) Audit help content and apply/revert updates. 3) Check the verified-watermark date in the Help modal. 4) Export your plan as JSON, saved with today's date — a permanent record of your plan as verified.



Every Help pane shows when its content was last verified against current law.

Chapter 37 — Troubleshooting and FAQ

Q: I open the file and see blank/strange formatting.
Use a modern browser (Chrome, Edge, Firefox or Safari from the last 2–3 years). Internet Explorer and very old Edge aren't supported.

Q: My plan disappeared after a Windows/browser update.
Clearing your browser's site data wipes the planner's local storage. Always export a JSON backup before a major OS or browser update, and re-import after.

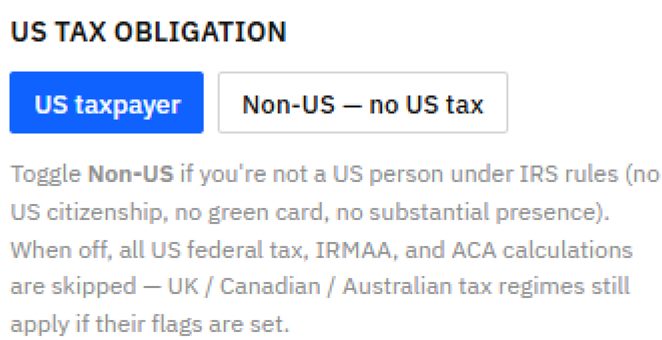
Q: I'm not a US citizen — does the planner still work for me?
Yes. Toggle *Non-US — no US tax* at the top of the Edit values tab. US federal tax, IRMAA, ACA and SS-provisional calculations zero out. UK, Canadian and Australian regimes still work.

Q: The Plan with AI sidebar keeps suggesting changes that run me out of money — why does it retry?
When you ask for a plan that doesn't deplete, the co-pilot tests each suggestion in the simulator and, if it still depletes, asks the AI to refine — up to three attempts. If it can't find a solvent plan, that's a signal your targets may be infeasible without bigger changes (later SS, lower spending, more conversions). See Chapter 20.

Q: Is my data — or my API key — uploaded anywhere?
Plan data stays in your browser's local storage. The only outbound calls are anonymous exchange-rate lookups and AI features you actively trigger (sent only to Anthropic's API, using your own key). Your key is stored only in your browser.

Q: Can I share it with my financial advisor?
Yes — export your plan as JSON or PDF. The `.html` file licence is single-user; for redistribution, contact WebNomad Studio.

Q: Is this financial advice?
No. The planner is an educational tool. Always verify your assumptions and consult a qualified professional before acting on any output.



Not a US person? One toggle removes all US-specific tax and healthcare modelling.

Appendix A — Keyboard shortcuts

The planner uses standard browser shortcuts:

- **Ctrl/Cmd + P** — open the print / PDF export dialog.
- **Ctrl/Cmd + F** — find text on the current page (useful in the Tax & ACA notes tab). **Esc**
- — close any open popover, modal, or the Plan with AI sidebar overlay on mobile. **Tab**
- **/ Shift+Tab** — navigate between input fields on the Edit values tab.

Appendix B — JSON export field reference

The plan JSON export contains a handful of top-level keys:

Key	Meaning
`S`	The full state object — every input value, including per-phase objects p1–p5 (now with a part-time income field in each phase) and state-tax settings.
`lumps`	Cash lump-sum objects, each with id, amtUSD, phase (1–5) and label.
`p5EndAge`	The age your plan runs to (default 80, often 90).
`activeCurrency`	Your active residency currency (USD / GBP / CAD / AUD / PHP / THB).
`subtitle`	Your scenario title (shown on the dashboard and PDF cover).
`annuityPlanState`	Records an applied annuity's premium, per-phase income additions and funding deductions for clean removal.
`learnDismissed`	Concepts-to-Learn entries you've dismissed.
`phAcknowledged`	Plan Health checks you've acknowledged (key, optional reason, timestamp).
`helpOverrides`	AI-applied audit corrections that survive reload and travel with the plan.
`v` / `date`	Format version and export timestamp — used for compatibility detection on import.

> **Note on AI settings**
> Your API key and the two chat histories (the AI Chat tab and the Plan with AI sidebar) are stored separately in browser local storage, *not* in the exported plan JSON — so sharing a plan never shares your key or your conversations.

Appendix C — Where to learn more

YouTube playlist

The WebNomad Studio YouTube playlist includes walkthrough videos covering the dashboard, first scenario setup, the Tax & ACA notes deep-dive, comparing scenarios, the Philippines expat scenario, the Roth conversion workflow, drawdown strategies, stress testing, and the AI advisor in practice. Search *"WebNomad Studio retirement planner"* on YouTube.

Where to buy

The Retirement Income Planner (this product) and the Roth Conversion Optimizer (Chapter 27) are sold directly at airetirementincomeplanner.com. A bundle of both is available at a discount.

WebNomad Studio

Website: airetirementincomeplanner.com · Email: dev@webnomad.org. Refined across seven major versions based on real user feedback from the retirement-planning community.

Retirement Income Planner — Version 7 User Guide · WebNomad Studio · webnomad.org · Published June 2026. This guide supersedes the v6 User Guide. Screenshot boxes are placeholders for the production layout.